SAKARYA UNIVERSITY JOURNAL OF COMPUTER AND INFORMATION SCIENCES
http://saucis.sakarya.edu.tr/
e-ISSN: 2636-8129
Publisher: Sakarya University
Vol. 9, No. 1, 262-291, 2026
DOI: 10.35377/saucis...1718848

REVIEW ARTICLE

From Traditional Methods to Artificial Intelligence: A Review of Non-Line-of-Sight Analysis Techniques

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Article History:
Received: 13.06.2025
Revised: 20.09.2025
Accepted: 18.11.2025
Published Online: 30.03.2026

ABSTRACT
This study explores the current state, core methodologies, and major challenges associated with non-line-of-sight (NLOS) sensing technologies. NLOS sensing enables the detection of objects and individuals outside the direct field of view and has critical applications in disaster response, security, and autonomous systems. Despite its growing potential, the field faces technical limitations, including restricted resolution, complex data processing, and multipath propagation effects. A wide range of approaches is examined, including both active and passive systems, SPAD and CCD/CMOS sensors, confocal and non-confocal imaging techniques, acoustic methods, and artificial intelligence-based models. The study also emphasizes innovative experimental setups and complex scene designs to evaluate system performance under realistic and challenging conditions. Furthermore, diverse evaluation metrics are discussed to support both numerical and perceptual analysis of system outputs. In conclusion, NLOS sensing is a complex field that requires an interdisciplinary approach, but it holds great potential for the scientific community and practitioners due to the opportunities it offers. This study has contributed to the current body of knowledge and provided suggestions that will guide future research.

Keywords: Non-line-of-sight (NLOS) analysis, Hidden object detection, SPAD and CCD/CMOS sensors, Confocal and non-confocal methods

1. Introduction

Non-line-of-sight (NLOS) analysis studies have made a breakthrough in the scientific world by providing information about objects hidden behind obstacles or walls. The first studies in this field were based on the theory of light interaction with surfaces, yielding impressive results, including the imaging of hidden scenes and the estimation of the motion and location of hidden objects. Today, non-line-of-sight analysis has applications in various fields, including autonomous vehicles, medical imaging, disaster response, and underwater research. Non-line-of-sight analysis, first introduced by Kirmani et al. [1] in 2009, was applied to real-world problems by Velten et al. [2] in 2012, utilizing femtosecond lasers and streak cameras. The sensor systems used in non-line-of-sight analysis studies, along with their technical specifications, play a crucial role in the obtained results. The high cost of traditional femtosecond lasers and streak cameras has led to the search for alternative systems. In this context, studies conducted using traditional cameras and single-pixel cameras, which are among the cost-effective devices [3, 4], also attract attention in this field. Additionally, many researchers preferred single-photon avalanche diodes (SPADs), which offer picosecond temporal resolution even in low-light conditions. In a study conducted using a SPAD sensor [5], the tracking of a hidden object at a distance of 1.43 km was successfully achieved through real-time measurement. This study demonstrated that NLOS analyses are applicable even at very long distances.

Non-visual field detection techniques are primarily classified into two categories: active and passive imaging. In addition to the sensor device used in active methods, the confocality of the light has a significant effect on the detection performance. While non-confocal methods enable the detection of larger areas, confocal methods offer the advantage of high resolution. Additionally, acoustic methods provide an innovative solution by utilizing sound waves, particularly in areas where light cannot penetrate. Passive methods, on the other hand, use only signals reflected from natural ambient light or light emitted by objects. These methods are low-cost and less affected by environmental effects.

One of the main problems of NLOS sensing technology is that it becomes difficult to obtain accurate data from signals that are scattered by obstacles. Depending on various environmental conditions, optical and acoustic signals encounter effects such as reflection, refraction and absorption, which directly affect the accuracy and sensitivity of such systems. However, the use of artificial intelligence (AI) approaches to solve these problems has significantly increased the efficiency of non-line-of-sight analysis. These methods can analyze more complex scenes in a shorter time and with higher accuracy compared to traditional approaches.

Cite as: S. Celebi, and I. Turkoglu, “From traditional methods to artificial intelligence: A review of non-line-of-sight analysis techniques,” *Sakarya University Journal of Computer and Information Sciences,* vol. 9, no. 1, pp. 140-169, 2026. doi: 10.35377/saucis...1718848

The criteria used in evaluating different non-line-of-sight analysis techniques play a crucial role in ensuring the comparability of methods in this field. These metrics enable the comparison of methods in terms of features such as accuracy, speed and cost, allowing the most appropriate technology to be selected for different application scenarios. There is no standard metric used in this field. The metric to be used is determined depending on factors such as the problem addressed and the proposed solution. For example, in an artificial intelligence-based study [6], accuracy and loss metrics were employed. In contrast, a physics-based imaging system utilizing event cameras [3] used the Peak Signal-to-Noise Ratio (PSNR) metric to compare the reference image and the reconstruction result.

In Non-Line-of-Sight (NLOS) analysis studies, scenarios based on the conventional three-bounce light transport model are commonly preferred. However, several applications that adopt unconventional approaches and explore novel scene configurations have also been reported in the literature [7, 8], [6]. The introduction of new scenes and scenarios contributes to expanding the applicability of NLOS analysis across a broader range of conditions. Additionally, data fusion approaches, which integrate information from various sensors and measurements, enable more accurate and comprehensive analysis of hidden scenes.

This study presents a comprehensive classification of non-line-of-sight (NLOS) sensing techniques, with a detailed examination of both active and passive methods. Core evaluation metrics and data fusion strategies are discussed, and the effectiveness and limitations of different approaches are identified. In addition, potential future research directions are highlighted, including novel scene types, scenarios, and methodologies, to foster innovative contributions to the field of NLOS analysis.

While recent surveys such as Geng et al. [9], which primarily examined conventional physics-based models, early stage deep learning approaches, and the introduction of new scene types, and Al-Ghafri et al. [10], which focused on NLOS communication technologies and the role of energy harvesting, have advanced the field, this review distinguishes itself by systematically integrating active, passive, and hybrid imaging paradigms. In particular, this study emphasizes underexplored dimensions such as multimodal data fusion (e.g., combining laser and acoustic signals), the adoption of transformer and self-supervised learning frameworks, and validation across diverse real-world scenarios, aspects that remain insufficiently addressed in prior reviews.

A closer examination of the literature further reveals that traditional methods such as Filtered Back Projection (FBP), Light Cone Transform (LCT) and Fourier-based Migration (FK) continue to be constrained by challenges such as resolution, multipath scattering, and hardware complexity, whereas AI and deep learning based approaches, though promising, face issues of generalization, dataset diversity, and hardware feasibility. By articulating these gaps, this review establishes a central thesis: advancing toward scalable and robust NLOS systems requires bridging the divide between traditional physics-based methods and AI-driven approaches through multimodal integration, diverse datasets, and real-world validation. In doing so, the study not only provides a theoretical and practical guide to the state of NLOS technologies but also offers a critical perspective and a roadmap for future research.

2. Classification of Non-Line-of-Sight Analysis Studies

In cases where a target is directly visible to a sensor or detector, Line-of-Sight (LOS) analysis is used, as illustrated in Figure 1(a). In this approach, the signal or light travels directly from the sensor to the target without encountering any obstruction, due to the presence of a clear line of sight between them. In contrast, Non-Line-of-Sight (NLOS) analysis is an advanced imaging technique that enables the detection of hidden or non-visible objects located behind walls or obstacles, as shown in Figure 1(b). Unlike conventional optical systems, this technique is based on analyzing light or acoustic signals that reach the hidden object via indirect reflections off surfaces and subsequently return through the same or alternate paths to the detector. The illuminated reflective surface redirects the signal toward the hidden object. Similarly, the signal reflected from the hidden object is redirected back to the reflective surface, and then to the original environment, where a detector captures it. This process, which involves sequential reflections from the reflective surface to the hidden object and back again, is referred to as the three-bounce transmission model. These three signal bounces are indicated in Figure 1(b) by red, black, and green arrows, respectively.

The three-bounce framework provides a mathematically tractable foundation and has been widely adopted in confocal and non-confocal NLOS configurations. However, each bounce introduces attenuation, multipath interference, and noise accumulation, which constrain resolution and limit the fidelity of reconstruction. Recent works have therefore refined and extended the classical three-bounce paradigm. For instance, Royo et al. [11] introduced “virtual mirrors” that exploit multi-bounce light transport beyond three interactions, while Pueyo-Ciutad et al. [12] incorporated polarization gating to address the missing cone problem in three-bounce NLOS imaging, thereby reducing directional ambiguities and enhancing robustness under complex scattering conditions. These advances show that although the three-bounce model remains central to NLOS imaging theory, it is increasingly viewed as a baseline to be adapted and extended for more realistic and challenging scenarios.

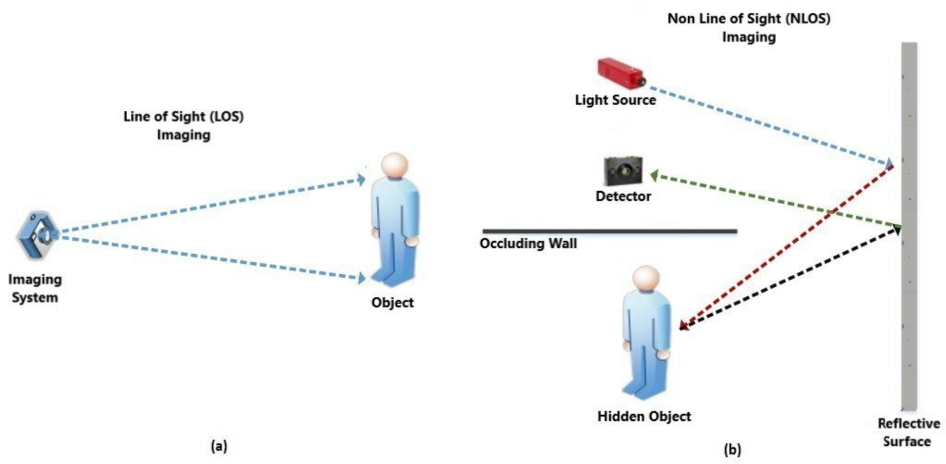

Figure 1. Basic Line-of-Sight (LOS) and Non-Line-of-Sight (NLOS) Analysis Diagram. LOS Involves Direct Signal Paths, Whereas NLOS Relies on Indirect Reflections from Relay Surfaces

NLOS studies can be categorized into three main groups based on their application and technical approaches, as illustrated in Figure 2: reconstruction algorithms, active methods, and passive methods. Active methods are classified as sensor-based methods, scanning-based methods, acoustic methods and artificial intelligence-based methods. Sensor-based methods include Single Photon Avalanche Diode (SPAD)-based and Charge Coupled Device (CCD)/Complementary Metal Oxide Semiconductor (CMOS)-based systems according to the sensor technologies used in NLOS studies. In this context, the term "sensor-based" is used to emphasize the underlying detector technology (SPAD, CCD/CMOS), even though these methods fundamentally rely on light detection. These sensors offer different advantages depending on their sensitivity to signal detection. Scanning-based methods are divided into two types: confocal and non-confocal systems, based on the location of the laser and detector, which directly affects the depth sensitivity of the scene scanning method. Acoustic systems are considered an alternative method to optical approaches and provide effective results in environments with insufficient light by providing NLOS scene information through sound waves. Artificial intelligence-based active methods are approaches that estimate invisible scenes from signals using deep learning models, providing high success rates, especially in complex and noisy data. Passive methods utilize natural ambient lighting, such as lamps, sunlight, and reflected light and shadows from objects, rather than a controllable external light source. Passive methods are divided into two groups: traditional techniques and AI-based techniques. Passive methods are less costly than active methods, but may result in lower-resolution reconstructions.

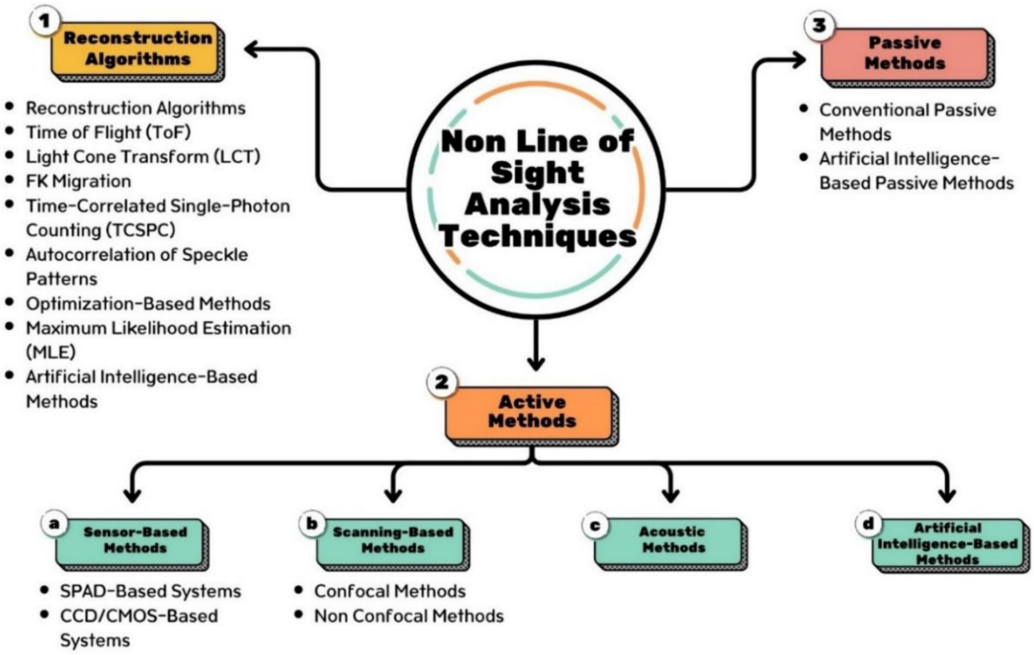

Figure 2. Classification of Non-Line-of-Sight Analysis Studies

In active and passive imaging, information received from sensors is processed with some algorithms to provide meaning to the hidden scene. Reconstruction algorithms are methods that enable the reconstruction of hidden scenes by processing data and are divided into subgroups such as Filtered Back Projection (FBP), Time of Flight (ToF), Light Cone Transform (LCT), Fourier/k-space (FK) Migration, Time Correlated Single Photon Counting (TCSPC), correlation of speckle patterns, optimization-based techniques, Maximum Likelihood Estimation (MLE) and artificial intelligence-based approaches. The workflow shown in Figure 3 comprehensively reflects the mechanisms of non-line-of-sight analysis methods discussed in this study.

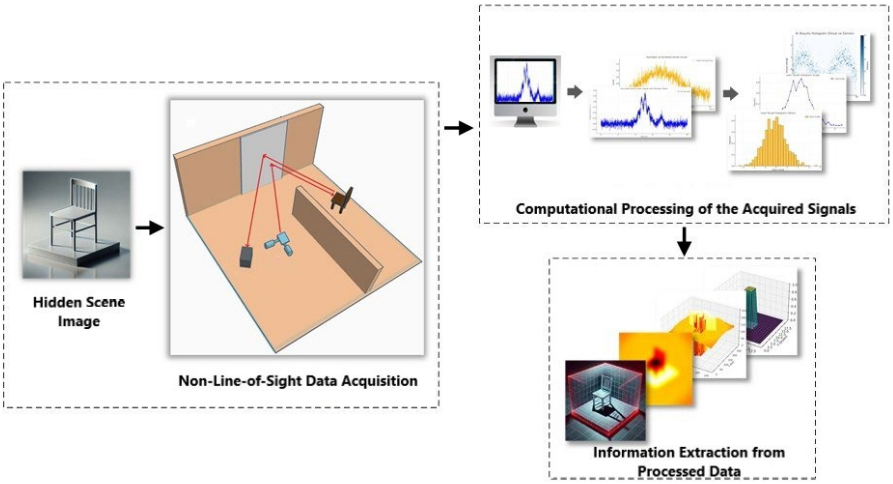

Figure 3. Non-Line-Of-Sight Analysis Process from Signal Propagation to Information Extraction

2.1. Reconstruction Algorithms

Non-Line-of-Sight analysis techniques comprise a range of methods developed to detect and accurately reconstruct objects that are located outside the direct line of sight. These techniques can be classified based on the type of signal used and the measurement strategy employed. Early studies primarily focused on fundamental mathematical approaches, such as backprojection. Over time, more sophisticated methods, including the light cone transform and FK Migration, as well as AI-based techniques, have been introduced. These methods rely on analyzing measurement data and generating a meaningful reconstruction of the hidden scene. Initial techniques gained popularity due to their low cost and ease of implementation. However, limitations such as low resolution and sensitivity to noise have led to increased interest in optimization-based and AI-driven approaches. Each technique presents its own set of advantages and disadvantages. For instance, Time-Correlated Single-Photon Counting (TCSPC) provides high temporal precision, while autocorrelation of speckle patterns offers a low-cost and simple setup. AI-based methods, on the other hand, deliver high-resolution reconstructions even in complex and highly variable scenarios. This diversity enables NLOS sensing to adapt to various application scenarios and to evolve continuously.

In this section, reconstruction techniques will be examined in chronological order according to their historical development.

Filtered Back Projection (FBP): This method is widely used, especially in areas such as medical imaging [13]. The basis of this technique is the reconstruction of the original image by reflecting the measured waves, such as sound and laser light. It works similarly in non-line-of-sight analysis, where a representation of the hidden scene is created from the back reflection information of sound and laser signals sent to objects outside the field of view. FBP consists of receiving projection data from various angles, filtering the data and back projection steps. First, signals are sent from different angles, and their reflections are collected by sensors, and projections are obtained with the Radon Transform. Each projection taken from different parts of the object provides indirect information about the object. The obtained projection data is converted to the frequency domain with the Fourier Transform in order to better analyze the frequency components. Filters are applied to project the image with higher accuracy and make it clearer. Since low-frequency components usually contain noise and high-frequency components provide more detail, high-pass filters are usually used. The filtered projections are then converted back to the time domain with an inverse Fourier transform. In the last stage, each projection of data is spread back equally to the image area. Each projection provides information for a limited region of the image, and when all projections are combined, the object's structure is more clearly represented. Improved FBP algorithms, such as error backprojection [14], Gaussian Laplacian FBP [15], and fast backprojection [16], have been proposed to provide non-line-of-sight reconstruction with higher resolution; however, these algorithms require significantly more computational power.

Time of Flight (ToF): This process involves calculating the travel time between a light or sound wave emitted from the source, scattering and interacting with the hidden object, and returning to the sensor, allowing for inferences such as distance measurement and depth maps. The hidden targets are reconstructed by determining the distance, shape and location of the reflected signal. The distance is determined basically as in Equation 1, using the time difference t between the return and propagation of the light.

$$\boldsymbol{d} = \frac{\boldsymbol{c} * \boldsymbol{t}}{\boldsymbol{2}} \tag{1}$$

Here:

d: Distance between light source and target.

c: Speed of light in the medium (approximately c = 3 x 108 m/s).

t: Flight time (total time for light to travel to and return from the target).

This method, which can be adapted to various wave types, such as optical, acoustic, and radar, ultimately measures the flight time with picosecond or nanosecond-level precision. For this reason, in systems using ToF calculation, the sensor must have superior detection sensitivity and high time resolution. The first system that demonstrated the practicality of ToF data acquisition in NLOS detection was developed using a line camera, which provided 2 ps timing sensitivity and 1 cm lateral spatial resolution [2, 17]. The most advanced non-line-of-sight analysis systems are also generally based on high-time-resolution detectors [18]. Although ToF measurements without line of sight can reconstruct hidden objects with centimeter precision, they require expensive equipment, such as single-photon detectors and nanosecond pulsed lasers, which can cost tens of thousands of dollars [6].

Light Cone Transform (LCT): It is a technique used especially in ToF based non-line-of-sight analysis, and its main goal is to represent the propagation and return paths of light with a cone-shaped geometric model and to provide reconstruction of the hidden scene. The propagation of light is represented as a "light cone". The light cone refers to the volume formed by the light hitting the surrounding surfaces following a conical path from the moment it is emitted from the source. In ToF non-line-of-sight analysis, a 5-dimensional dataset is recorded: 2D surface coordinates scanned by the laser and 3D time-measured information of the light for each position (x, y, t). Reconstruction is provided by a back-projection algorithm, but it has $O(N^6)$ complexity (N = scene resolution), and the processing time increases rapidly with increasing resolution. By combining laser illumination and photon detection at the same point, the 5D dataset is reduced to 3D (x, y, t) and the ToF data is convolved with a kernel representing the propagation and return paths of the light (light cone) to obtain the information of the hidden scene. The hidden scene is clarified through the deconvolution process, and noise is eliminated using the Wiener Filter. The ToF data and light cone kernel are transferred to the frequency domain using the Fourier transform, which accelerates the convolution process. The 3D scene is then reconstructed by transferring it back to the time domain using the inverse Fourier transform. Thus, the processing complexity is reduced to $O(N^3 \log N)$, and a high-resolution 3D scene can be created in milliseconds [19].

FK Migration: It is a reconstruction method used in non-line of sight analysis, used in confocal applications such as LCT and is based on 3D convolutions. In non-line-of-sight analysis, the data recorded by the sensors contain temporal and spatial information. To better analyze the wave motions, the recorded data are first transformed into the frequency (f) and wave number (k) domains, and then different surface reflections are modeled using wave propagation equations. These models simulate propagation by calculating wave propagation paths, speed, and direction. Thus, the incorrectly positioned reflection data are determined and moved to their original positions according to the propagation model. Finally, the corrected data is transformed into the space-time domain by the Inverse Fourier Transform. The main limitations of the method are as follows [20]:

- Since it can only use the light reflected from the confocal position on the relay wall, most of the light cannot be utilized.
- Although it is suitable for real-time scanning with retro-reflective objects, scanning time increases on randomly distributed surfaces.
- Although it provides fast results for low-resolution reflected scenes, high memory usage and efficient confocal capture are required for complex scenes.

Coherence-based methods: These methods exploit the spatial or temporal coherence of scattered light to reconstruct hidden scenes. Unlike time-of-flight approaches that depend on direct travel-time measurements, coherence-based techniques analyze the statistical correlations and phase relationships within the scattered wavefront, by studying how the coherence of the light field changes after interacting with hidden objects and reflecting from visible relay surfaces, structural and positional information can be inferred even when intensity-only measurements are insufficient. This principle allows reconstruction without requiring ultrafast detectors, as coherence functions can often be measured with standard imaging sensors. As a result, coherence-based methods provide a cost-effective and flexible alternative, especially in situations where active illumination is impractical or expensive. They thus represent a complementary strategy that broadens the range of modalities available for non-line-of-sight scene reconstruction.

Time-Correlated Single Photon Counting (TCSPC): This method is used to precisely measure the time information related to the reflection of light from surfaces. This method usually requires pulsed laser sources operating at specific time intervals and sensors with TCSPC modules. Short-duration laser pulses are sent at specific time intervals with the pulsed laser source. In this way, the timing of the laser signals can be precisely controlled. The signals travel from and back to the hidden scene, resulting in a specific time delay depending on the properties of the objects in the hidden scene. The arrival time of each photon in the laser signals returning to the sensors is recorded with picosecond precision. The TCSPC method calculates the time delay between the laser pulse and the sensor detection, and these measurements are collected in a histogram. The time delay and the density of the photons allow information about the distance, geometry and location of the hidden object to be extracted.

Speckle Pattern Autocorrelation: When a coherent light source (i.e., a laser with stable phase relations) hits a rough surface, the light reflected from the surface creates speckle patterns consisting of bright and dark dots. In non-line-of-sight analysis, speckle patterns created by light reflected from a hidden object hold hidden information about the shape, size, and motion of the object. Speckle patterns on the reflection wall are usually recorded with the help of conventional cameras. The similarities in intensity across different parts of the recorded patterns are measured using autocorrelation calculations to infer the surface and dimensions of the object. Moving objects create speckle patterns that change over time, and the motion of the object can be detected by analyzing the temporal autocorrelation of these patterns. This method can be implemented using low-cost devices, provides effective results even in low-light conditions, and is computationally efficient.

Optimization Based Methods: Optimization-based methods in non-line-of-sight analysis are an effective approach used to reconstruct the hidden scene from data obtained through scattering and indirect reflections. These processes formulate an inverse problem by relating the motion of light in the scene to physical and mathematical models. In the solution process, optimization algorithms are used to minimize the error between the measurement data and the scene estimation. These algorithms provide reconstruction with scene details such as light dynamics, reflection angle and surface properties. In addition, regularization terms such as L1 norm and L2 norm are used to reduce the uncertainties and noise effects that arise in the inverse problem solution. Optimization-based methods stand out with their high accuracy, especially in cases where data is limited and in complex scenes. However, the computationally intensive nature and long processing time of these methods usually limit their application areas to advanced hardware and software infrastructures.

Maximum Likelihood Estimation (MLE): This is a statistical approach widely used in non-line-of-sight analysis applications to determine the most probable parameters that are compatible with the observed data. This method first develops a probability model that describes the physical properties of the scene. A probability function representing the relationship between the measured data and this model is then created, and the logarithm of this function is maximized. This optimization process enables the determination of the most suitable parameters and the accurate reconstruction of non-line-of-sight scenes or reflective surfaces. MLE is robust to noise and can work in harmony with various physical models, enabling accurate results. In this way, NLOS scenes can be imaged more clearly and accurately using data such as time-of-flight measurements.

Artificial Intelligence-Based Methods: These are an effective approach to solving the problem of non-line-of-sight analysis. These methods learn complex patterns to understand light reflection, refraction, and distribution by learning with large datasets. Deep learning models such as convolutional neural networks (CNNs) extract local features in the data and make accurate predictions from noisy and nonlinear NLOS data. Neural networks trained on large datasets offer a fast and effective reconstruction process by modeling the complex, nonlinear relationship between physical measurements and the target object [8]. These approaches offer a significant advantage in non-line-of-sight analysis, yielding better results than traditional methods.

2.2. Active Methods

Active NLOS imaging methods are examined under four main headings: sensor technology, scanning strategy, alternative physical approaches, and artificial intelligence-based analyses. Sensor-based methods are classified according to the types of detectors used. In this context, high-sensitivity Single Photon Avalanche Diode (SPAD) sensors and, more commonly, yet economical, CCD/CMOS sensors stand out. These sensors have different performance characteristics in terms of photon detection capacity and time resolution. Scanning-based approaches are divided into confocal and non-confocal systems according to the location of the laser and detector; this structure directly affects how the scene is scanned and the accuracy of the depth data obtained. Acoustic systems, considered as an alternative to optical methods, offer a solution for detecting objects outside the field of view by using sound waves and provide advantages, especially in environments where light is limited. Artificial intelligence-supported methods, on the other hand, make it possible to estimate invisible scenes from low-quality signals through deep learning models and produce effective results, especially on complex data.

2.2.1. Sensor Based Approaches

2.2.1.1. SPAD Based Active Methods

Single Photon Avalanche Diodes (SPAD) are sensors that record the light transmission and optical response of a scene under transient lighting conditions and analyze them in relation to time. These sensors, triggered by photon transmission, can operate even in low light conditions and offer picosecond temporal resolution. SPAD sensors calculate the return time of the photon that is sent and bounces off surfaces along indirect paths, thus obtaining depth information. The photon hits the photodetector region of the SPAD, and when a critical voltage level called the breakdown voltage is reached, a self-increasing macroscopic current occurs. The energy of the photon is transferred to an electron in the atoms of the semiconductor photodetector region. The electron that gains energy leaves the orbit and becomes mobile, and a hole is formed where the electron was. The light signal is converted into an electrical signal with the formation of an electron-hole pair. SPAD sensors operate under a high voltage (bias voltage) in the Geiger mode. This high voltage increases the electric field inside the device (>3×105 V/cm) and causes even a single electron to cause an avalanche effect [21]. The avalanche effect causes a single charge carrier reaching the active region of the SPAD to trigger approximately 108 carrier particles [22]. The release of the electron and the start of the avalanche reaction indicate the photon arrival time. The avalanche effect continues until the current drops below a certain voltage (breakdown voltage). The quenching circuit [23] stops the avalanche effect by reducing the voltage, and the photodiode is reset, and the bias voltage is recharged to detect another photon. This process is repeated for each photon detection. Some special electronic circuits are required for the operation of SPADs. These circuits perform the following operations [21]:

i. Detect the start of the avalanche current.

ii. Produce an output signal at the same time as this detection and report the arrival of a photon.

iii. Reduce the voltage of the device to extinguish the avalanche effect.

iv. Return the device to the starting voltage to detect a new photon.

Several parameters affect the performance of SPADs. These factors affect the efficiency, sensitivity, and accuracy of the SPAD's photon detection and should be optimized according to the sensor's specific usage areas. Timing jitter refers to the error margin of a detector in the time it takes to detect a photon. It can be expressed as the detection of photons sent under the same conditions and at the same time t at different moments with small time differences. In other words, they are the time deviations that occur during the measurement. Timing resolution refers to the sensor's sensitivity to detect the arrival time of photons. High-resolution SPADs can detect even very small time differences between photons. Signal to Noise Ratio (SNR) is a metric that expresses the ratio of the signal strength detected by the sensor to unwanted noise. Dark Count Rate (DCR) is the incorrect counts made by a sensor that does not receive any photons (is not exposed to light) due to noise. Thermal energy, circuit elements and external noise in the sensor can cause the sensor to trigger incorrectly by creating electron-hole pairs even in an environment where there are no photons. Especially in applications where weak signals are used, low DCR provides more accurate data measurements. DCR and SNR values change inversely. Afterpulsing is the process of releasing the residual charge carriers from the previous measurement after the avalanche effect is damped and the detector is reset, initiating a new avalanche process. A signal is generated even though no real photon is detected, which reduces the counting accuracy and negatively affects the system performance. Photon Detection Efficiency (PDE) expresses the probability of detecting photons that hit the detector and converting them into a signal. The detector's photon detection and avalanche initiation success directly increases efficiency. High PDE ensures that noise is minimized and performance is increased. Recovery Time is the time required for the SPAD to return to its initial position and for the bias voltage to be recharged, allowing it to detect a new photon.

SPAD detectors are divided into two groups according to the structure of the depletion layer: thin and thick SPADs. The depletion layer is the active SPAD area where the photon is first detected, and the avalanche effect is triggered. The structure of this area is one of the important factors affecting the sensitivity and response speed of the SPAD. Thin SPADs generally have a 1 μm thick depletion layer and a small active area (20 μm to 100 μm in diameter). In comparison, thick SPADs have a depletion layer between 20 μm and 150 μm thick and a larger active area (100 μm to 500 μm in diameter) [21]. While thin depletion layers provide fast response times, thick layers can withstand higher energy, which increases the response times. While the breakdown voltage for thin SPADs is approximately 30-35 V, this value has been measured to be 300-500 V for thick SPADs [24]. A low breakdown voltage offers power consumption efficiency for thin SPADs; however, the small active area and this voltage level may impose limitations on sensitive measurements, such as long-distance sensing or imaging at low light levels. In thick SPADs, high breakdown voltage and active area provide high avalanche current and sensitivity; however, due to this avalanche current and high voltage, more power consumption and heat generation occur. The timing resolution for thin SPADs can be as low as 20 ps, while this value is approximately 150 ps for thick SPADs. Full Width at Half Maximum (FWHM) can be as low as 30 picoseconds for thin SPADs, while this value can be as high as several hundred picoseconds for thick SPADs [24]. This provides faster and more sensitive photon detection in thin SPADs. On the other hand, the low dithering in thin SPADs leads to lower photon detection efficiency in the red region (17% at 800 nm), while thick SPADs have higher detection efficiency in the red and near-infrared region (60% at 800 nm) [25]. Table 1 compares the properties of thin and thick SPADs.

Table 1. Properties of Thin and Thick SPADs

Feature	Thin SPADs	Thick SPADs
Depletion Layer Thickness	Typically 1 μm	20–150 μm
Active Area Diameter	20–100 μm	100–500 μm

Table 1 (Continued)

Breakdown Voltage	30–35 V	300–500 V
Timing Resolution	20 ps	150 ps
Timing Jitter	Very low (30 ps)	High (several hundred ps)
Photon Detection Efficiency (PDE)	Low, especially at long wavelengths (~17%)	High in red and near-infrared regions (~60%)
Applications	Applications requiring fast timing	Red/NIR applications requiring high PDE

SPADs are divided into two groups according to their material structures: normal SPAD (Si-SPAD) and InGaAS/InP SPAD. SPADs operating in the visible light spectrum are generally silicon-based and are called Si-SPAD. These sensors are used in non-line-of-sight analysis systems operating with visible laser sources; however, since they operate with visible light, other light sources in the environment, such as sunlight and lamps, can affect their performance. InGaAS/InP SPAD is produced using Indium Gallium Arsenide (InGaAs) and Indium Phosphide (InP) materials. The first study on InGaAs/InP SPAD was carried out by Levine and Bethea in 1984 [26]. These sensors operate in the infrared spectrum with invisible light sources. Since infrared light scatters less, it can overcome obstacles and measure at longer distances, making it perform better in NLOS applications. InGaAs SPAD is a device frequently used in fields such as quantum communication and quantum key distribution [27], 3D LADAR imaging, high energy physics [28] and fluorescence lifetime imaging [29], operating in the near-infrared wavelength range between 1.0 µm and 1.7 µm. While Si-based SPADs have a cut-off wavelength below 1.55 µm, InGaAsP-based SPADs have a wider wavelength range [30]. InGaAs/InP SPAD provides superior performance and sensitivity but generally has a higher cost and cooling requirements. In contrast, Si SPAD is preferred especially in low-budget projects, offering a more economical and feasible alternative. InGaAs/InP SPAD detectors offer 150 ps time jitter while operating in the wavelength range of 850-1550 nm with 20% quantum efficiency and a dark count rate of less than 200 Hz. In contrast, the Si SPAD detector offers 50% quantum efficiency, 10 kHz dark count rate, and 200 ps time jitter with a wavelength range of 320–900 nm.

InGaAs SPADs are often operated in off mode due to their after-pulse problems and limited bandwidth, while free-running InGaAs/InP SPADs are preferred for applications requiring random photon flow; however, such SPADs are subject to the after-pulse effect due to carrier traps during avalanche events, which can be mitigated by active or passive quenching techniques [31]. The detection efficiency varies between 10% and 30%, while the dark count probability per gate is in the order of 10^{-6} to 10^{-4} [32]. The photon detection efficiency (PDE) and dark count ratio (DCR) in InGaAs/InP SPAD, which are critical for photon counting communication, depend on fundamental properties that cannot be improved by peripheral circuits. An integrated SPAD model that combines these parameters is developed, and a bit error rate (BER) model is generated that provides more accurate estimates by considering non-ideal factors, thereby optimizing the BER in photon counting communication by examining the design elements [30]. Although InGaAs/InP SPADs are compatible with optical fiber communication, offer compact size, and benefit from Peltier cooling, they suffer from high avalanche event rates [33].

SPAD detectors can be configured as single detectors or multi-pixel arrays to analyze different types of optical signals using three different methods: Single Photon Counting (SPC), photon timing, and time-gated SPC [34]. The SPC method is a technique for counting signals with very low photon counts or signals in environments with low light levels at the microsecond level, utilizing the ability of SPAD sensors to detect photons individually. With this method, precise measurements can be made at low light levels, and the intensity and time-dependent change of the optical signal can be measured precisely. Photon timing is performed by recording the arrival times of the photons that are detected. A detailed analysis of rapidly changing optical signals in the context of time can be performed with TCSPC. Time-gated single photon counting enables the signal to be analyzed by precisely limiting it with short-term measurement intervals, known as integration windows. Different points of the signal can be examined by shifting the opening and closing times of the integration window, and the temporal profile of the signal can be obtained by combining them. It is possible to make much more precise measurements by narrowing the time window. To analyze the signal effectively with the time-gated SPC method, Time-to-Digital Converter (TDC) devices are integrated. With TDCs, the arrival time of the photon is converted to digital, and measurement is possible at microsecond or nanosecond levels. In this way, the timestamps of the photons within the time window are created, and the time profile of the signal is precisely reconstructed.

In order to increase the detection area and data collection speed of single detector SPADs, studies have been carried out since the end of the 20th century on the development of multiple SPAD chips [34]. SPAD arrays used in commercial LiDAR systems are formed by combining multiple SPAD sensors, which saves scanning time. They provide parallel detection by simultaneously detecting photons from multiple points, thus enabling very fast data collection. While the field of view of single detector SPADs is limited, SPAD arrays can scan large areas with high resolution. Each detector in the array detects photons coming from different regions, and high-resolution data is obtained by combining these photons.

SPAD sensors, which are used in many areas such as fluorescence lifetime measurements, time-resolved Near Infrared Spectroscopy (NIRS), and quantum cryptography [35], stand out in non-line-of-sight analysis with their single-photon detection capability and picosecond time resolution in addition to their low cost and compact structure [36]. SPAD sensor-based active NLOS imaging systems offer effective performance, especially in low-light conditions and long-distance scenarios, thanks to their high time resolution and single-photon detection capabilities. Zappa et al. [21] investigated the crosstalk effect in adaptive optics systems using a monolithic 240-pixel SPAD array. In another system developed in 2015 [37], imaging was performed at a range of 1 meter for applications such as security and surveillance with a 32×32 resolution SPAD camera operating at a wavelength of 800 nm. In more advanced systems, reconstruction-based methods have been developed using single-pixel SPAD sensors and laser sources guided by galvanometer mirrors [38]. In systems equipped with high-power pulsed lasers and TCSPC modules [39], detection up to 53 meters at a wavelength of 1532 nm was possible. In addition, a system proposed in 2020 [36] performed 3D imaging at a range of 1.2 meters with a scanner-free setup. In another study conducted in the same year [35], timing accuracy and crosstalk analysis were performed using gated SPAD sensors in a 16x1 array; time resolution was focused on with a 24-hour data collection period. In a system developed in 2021 [40], NLOS detection was performed using a 256-pixel (16×16) SPAD array with an increased filling ratio with a microlens array. In another study [41], the detection of hidden objects at a distance of 1.25 meters was tested using SPAD arrays with different timing resolutions. Both real SPAD arrays and 64×64 simulation data were utilized in this system. In recent years, resolution and timing accuracy have increased significantly. For example, a system developed in 2024 [42] performed motion detection and depth mapping at 500 frames per second with a 128×128 pixel SPAD array. In one of the most advanced examples [5], hidden object tracking was achieved at a range of 1.43 kilometers, thanks to real-time imaging provided for urban environments using InGaAs/InP SPAD detectors and a dual-telescope confocal optical design. Finally, a SPAD system operating at a wavelength of 940 nm [43] was developed for depth mapping and material classification up to 4.7 meters; this system was supported by galvo mirrors, optical filters and focusing lenses. These examples show that SPAD-based systems can be adapted quite flexibly at different wavelengths, resolution levels and optical arrangements. Although the sensitivity, range and purpose of use of each system are different, SPAD sensors have generally become a powerful tool that offers high success in non-line-of-sight analysis. Table 2 provides an overview of NLOS systems developed using SPAD cameras.

Table 2. Properties of Thin and Thick SPADs

Publication	Year	Wavelength	Range	Temporal Resolution	Application	Sensor	Optical Setup
[21]	2007	-	-	-	Astronomy, adaptive optics (AO)	240 px (60 quad-cells with 20–75 μm)	Pixels spaced at 1 mm, deep isolation
[37]	2015	800 nm	1 m	~50 ps	Security, surveillance	32×32 px	Reflective ground, SPAD camera
[38]	2015	1030 nm (laser), 515 nm (converted)	1 m	30 ps	Reconstruction, laser projection	Single pixel	Galvo mirror, SPAD detector
[39]	2017	1532 nm	53 m	~200 ps	NLOS detection, security, medical, robotics, object tracking	Single pixel	Pulsed laser diode, SPAD detector, TCSPC, telescope
[36]	2020	532 nm	1.2 m	165 ps	Scannerless NLOS 3D imaging	32×32 px	SPAD camera, 70 ps laser
[35]	2020	850 nm	-	42 ps	NLOS, crosstalk reduction, timing studies	16×1 array	Fast-gated SPAD, electroluminescence measurements
[40]	2021	820 nm	-	6 ps (TDC)	NLOS detection	16×16 px (256)	Microlens array, 9.6% fill factor, up to 80% enhancement
[5]	2021	1550.2 nm (near-IR)	1.43 km	32 ps	Long-range NLOS, urban tracking	High resolution (14 cm)	Confocal design, dual telescopes, InGaAs/InP SPAD

Table 2. (Continued)

[41]	2024	-	1.25 m	60/57/20/ 10 ps	NLOS with low-resolution SPAD	32×32 (real), 64×64 (simulated)	Relay wall, pulsed laser, SPAD array
[42]	2024	483 nm	3.5 m	5 ps (step size)	3D imaging, motion, depth mapping	128×128 px	Picoquant pulsed laser, 40° FOV, 800 nm filter, optical lens
[43]	2024	940 nm	4.7 m	1.3 ns	ToF, NLOS object tracking, material classification	16×16 px	Galvo mirror, filter, lens-focused sensor

2.2.1.2. CCD and CMOS Based Active Methods

Charge-Coupled Device (CCD) and Complementary Metal-Oxide-Semiconductor (CMOS) sensors are used to obtain images by converting light intensity into digital signals. Incoming light excites pixel arrays on these sensors, and each pixel measures the light intensity, converting it into an analog signal. Photodiodes are used to convert light into electrical signals.

Sharma stated that Willard Boyle and George Smith invented CCD sensors at Bell Laboratories in the USA in 1970 [44]. CCD sensors transfer electrical charges generated from light sequentially from pixel arrays. The signal generated by each pixel is transferred to the photodiode of the neighboring pixel, and finally, all the transmitted signals are collected in the reading area of the sensor. The collected data is converted into digital data by an analog-digital converter (ADC). The sequential nature of electrical transmission causes the data reading process to be slow and consumes more energy. High energy consumption is a limiting factor for mobile systems. On the other hand, data is read with high sensitivity, and the noise ratio is low. These sensors are utilized in areas such as low-light photography, as they offer excellent performance in low-light conditions. High performance in low-light conditions is quite advantageous in NLOS detection because the important point in NLOS detection is to detect weakly reflected light with low noise and high resolution. CCD sensors can precisely measure the intensity of reflected light, the angle of incidence and the interaction between surfaces. Thus, the dynamic properties of light, as well as the shape and movement of the hidden object, can be analyzed. The complex structure and high cost of CCD sensors can be disadvantages for NLOS detection applications.

In CMOS sensors, each pixel consists of a photodiode and an independent signal amplifier circuit. The signal obtained in each pixel is processed separately and converted into digital data via an ADC. The independent processing of pixels allows data to be received in parallel. Thus, the data reading process is completed faster and with lower energy consumption compared to CCD sensors. Due to the amplifiers, the noise ratio is higher compared to CCD sensors. Since it is low-cost, it is used in consumer electronics and applications requiring high speed. In non-line-of-sight analysis applications, it is used in intensity-based imaging studies, such as those employing CCD sensors. However, it is preferred over CCD sensors in dynamic systems where data must be collected quickly. It requires stronger light compared to CCD sensors for use in NLOS detection applications. The resolution of these sensors is determined by the number of pixels of the chip used. Current CMOS sensors can provide a resolution of up to 20 Megapixels and can measure the depth of objects in detail with a depth sensitivity of 330 μm [30]. This demonstrates that these sensors are capable of performing precise 3D measurements in NLOS detection studies and provide clear, detailed images.

Lei et al. [6] propose an NLOS imaging method that can identify objects outside the field of view by analyzing speckle patterns emitted from a surface illuminated by a coherent laser source (a source with fixed phase relations enabling interference-based measurements). In this system, a CMOS image sensor with a cost of only a few thousand dollars and a laser source were used, and direct object and human pose recognition operations were performed without the need for images with the help of deep learning algorithms. This method, which achieved a success rate of over 90%, has technical advantages such as low hardware complexity and compatibility with different wall types. However, it is sensitive to difficulties such as vibration-induced noise, low signal-to-noise ratio (SNR), and limited distance. Similarly, Ballester et al. [45] developed an interferometric ToF camera using the native resolutions of CCD and CMOS sensors, enabling 3D measurement of dynamic and rough surfaces with submillimeter precision in a single shot. This system can record videos of moving objects and create dynamic 3D models with high resolution (>2 Mp point cloud) and offers a wide range of applications from augmented reality to medical imaging. In addition, thanks to the synthetic wavelengths created by the simultaneous use of multiple lasers, more precise depth maps can be obtained; thus, approximately 100 times higher sensitivity is achieved compared to classical ToF systems. Despite being able to operate at high resolution, time resolution is limited by the detector response time. In this direction, CCD and CMOS sensor-based systems offer significant advantages in terms of both cost-effectiveness and resolution in NLOS imaging; however, they are evaluated with some limitations, such as time resolution.

2.2.2. Scanning-Based Techniques

2.2.2.1. Confocal and Non-Confocal Methods

Non-line-of-sight detection systems generally rely on time-resolution detectors; however, the use of these detectors brings disadvantages, including high computational and memory requirements, weak light flux, and expensive hardware requirements [18]. Non-line-of-sight detection processes time-of-flight data to create a 3D model of the hidden scene. In this process, the scattering of the signal generates large data sets, which necessitate complex mathematical calculations for analysis. The energy of the signals sent to the hidden scene and scattered decreases as they interact with the surfaces, causing the signal that returns and reaches the detectors to be extremely weak. In addition, time-resolution detectors are usually high-cost. To overcome these challenges, O'Toole and his team [18] derived the LCT model in 2018 by focusing the laser light and detector at the same point, thus laying the foundation for confocal methods. Confocal methods are systems in which the light or sound source and the sensor detector are positioned along the same optical axis; thus, the emitted and returned signals are measured from the same direction, as shown in Figure 4 (a). The LCT model facilitates the reconstruction of the hidden scene using time-of-flight data obtained by reflecting laser light from surfaces with a specific geometry, and constitutes one of the most important advantages of confocal systems. In contrast, in non-confocal methods, the source and detector are positioned along different axes, as illustrated in Figure 4 (b). Beyond this fundamental geometric difference, non-confocal methods generally collect time-resolved data by simultaneously illuminating different points of the visible surface rather than scanning the surface.

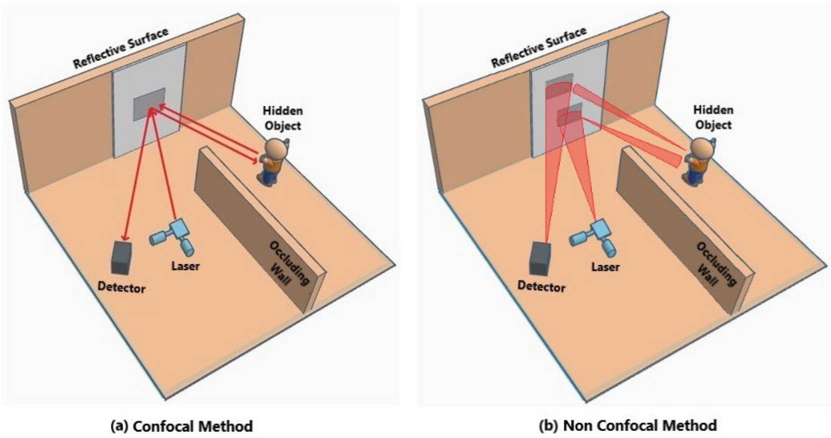

Figure 4. Confocal and Non-Confocal NLOS Configurations Schematically Illustrated for This Review. (a) Confocal Method: The Light Source and Detector Are Positioned on the Same Axis, and Signals Are Collected from a Single Point. (b) Non-Confocal Method: The Source and Detector Are Positioned on Different Axes, and Signals Are Collected Simultaneously from a Wide Area with a Single-Shot Illumination.

In ToF NLOS detection, a 5-dimensional dataset is obtained: 2D surface coordinates scanned by the laser and 3D time information of the light for each position (x, y, t). In confocal methods, this dataset is reduced to 3D (x, y, t) by converging the laser light and the detector at the same point. This reduction process can be made more efficient by methods such as the Light-Cone Transform (LCT). LCT tries to obtain scene information by convolving ToF data with a kernel representing the propagation and return paths of the light. However, similar operations for non-confocal methods become more complex and usually require more processing power. Although confocal methods provide higher resolution and faster scene reconstruction, there may be difficulties, such as the high cost of the equipment used and the sensitivity of the system.

The advantages of confocal methods can be listed as follows:

- Since the light source and detector measure at the same point, the signal-to-noise ratio in the target region increases and the data size is reduced. Thus, the processing time is shortened, and the memory requirement is reduced.
- Since the use of scattered light is reduced, more detailed 3D images are obtained.
- While the reconstruction of the hidden scene is performed faster and more efficiently thanks to mathematical models such as the light-cone transform and operations such as the Fourier transform, noise is reduced with the Wiener Filter.

The disadvantages of confocal methods are as follows:

- The laser must be moved to scan a wide area. This requires a complex scanning mechanism and increases the measurement time.

- Simultaneous, precise measurements may require costly hardware.
- Scanning is difficult for dynamic scenes and may produce erroneous results.

Non-confocal methods, on the other hand, generally adopt the approach of simultaneous illumination of different points of the visible surface instead of scanning the surface and collecting time-resolution data. Instead of a single signal pulse, multiple signal sources are sent to the visible surface simultaneously. A series of detectors simultaneously records the signals scattered from the surfaces. While confocal methods require raster scanning to obtain data from a wide area, non-confocal methods collect wide-area data in parallel, thereby reducing processing time considerably. In non-confocal systems, information about the hidden scene is obtained by calculating the round-trip times of the signal. This data is processed with approaches such as back-projection algorithms, optimization-based methods or deep learning algorithms to reconstruct the 3D structure of the hidden scene.

The advantages of non-confocal methods can be listed as follows:

- Instead of raster scanning the surface, the entire scene is measured simultaneously, reducing time and hardware complexity.
- Reconstruction of dynamic scenes can be achieved using detectors such as SPAD cameras.
- Since it does not require a scanning mechanism, the system can be more compact and economical.
- Data collection can be accelerated by collecting data simultaneously from all pixels without scanning.

Disadvantages of non-confocal methods are as follows:

- Increased use of scattered light and scattering can lead to lower sensitivity and accuracy.
- Requires more memory and computational power as more data is collected from the scene.
- Ambient light affects the efficiency of the system, and the signal-to-noise ratio can be lower than that of confocal systems.

The study by Lindell et al. [46] has made significant progress in NLOS reconstruction with the F-k Migration technique using the confocal method and SPAD sensor with high-power lasers (>1W). Although the method requires a long exposure time (180 minutes), it offers a fast calculation opportunity with a processing time of 1.2 seconds in dynamic scenes. Its robustness against noise and high performance in different conditions, such as diffuse and specular reflection, are the main advantages of the method. It is also noteworthy that it does not require iterative calculation and produces successful results even in complex reflection conditions. However, there are limitations, such as neglecting the phase effect, low modeling success of wave fields at high frequencies, and security risks associated with high-power lasers. Poor time resolution at low signal levels and a limited coverage area are among the other disadvantages; however, the error rate is 1.8-3.5% lower compared to other methods, thereby increasing the reliability of the method.

The study by Jin et al. [36] provides successful object reconstruction using a non-confocal method using a 32x32 SPAD camera and a scanner-free NLOS 3D imaging system. The system, which works with a 70 ps pulsed erbium fiber laser, stands out with its scanner-free, compact structure, low power consumption and fast frame rate. The method, which provides high resolution and stability, is quite successful in terms of object reconstruction, but has disadvantages such as low fill rate (2%), time delay and the need for calibration of dark count rate differences. These limitations result in a positional error of approximately 41 cm. ToF measurements can be completed in 3 seconds.

The work of Manna et al. [7] provides the opportunity to collect data from both static and moving surfaces using silicon SPAD and TCSPC mode with a non-confocal method. The system, which processes data at picosecond resolution using a Onefive Katana 10 HP fiber laser, offers a strong signal-to-noise ratio and high success even on moving surfaces with the phasor domain reconstruction method. However, data loss and laser guidance difficulties on moving surfaces are among the main limitations of the method. The processing time is completed in a short time of a few minutes, and the error rate is low.

The work of Liu et al. [20] presents an important method in the non-line-of-sight analysis using SPAD sensors and a discrete phasor field diffraction model with a non-confocal approach. This method, based on Fast Fourier Transform and inverse diffraction techniques, is characterized by low memory usage and efficient processing times, while providing high accuracy rates. However, it has limitations such as increased spatial distortions and the need for depth calibration compared to confocal methods. The method has advantages over similar techniques in terms of processing time.

The work of Nam et al. [47] aims to successfully reconstruct real-time NLOS videos of reflective objects using only 28 SPAD pixels. The laser system used (Onefive Katana HP laser) and specially designed SPAD arrays, together with fast reconstruction algorithms, improved non-line-of-sight detection performance; however, it faced problems such as the long duration of confocal measurements, light losses, and signal delays. This method also enables the comparison of confocal and non-confocal measurements, particularly in overcoming difficulties such as light losses and noise.

The study by Wang et al. [48] aims to perform NLOS detection with SPAD technology and picosecond temporal resolution. With the 1.4 ps resolution obtained using a picosecond pulsed laser, high depth resolution and 3D information were provided, yielding clear details. With the LCT method, successful results were obtained, featuring an axial resolution of 180 μm and an approximate lateral resolution of 2 mm. However, difficulties were encountered, such as Lambertian surfaces causing letters to be divided and letters becoming unreadable in images obtained with 5 ps resolution. The study was conducted efficiently, with 1,000 scan times in each 50-ms time slot.

Zhang et al.'s work [49] uses SPAD (PD-100-CTD) technology and the Archimedean spiral scanning method to perform confocal NLOS detection. With the 670 nm pulsed laser (ALPHALAS) and a global regularization method featuring a 4 ps time resolution, the need for multi-point raster scanning is significantly reduced, thereby shortening the data acquisition time. This approach saves time for moving objects, while the imaging quality is observed to be slightly lower than that of the multi-point raster scanning method. However, the method used is much faster, as approximately 1/23 of the scanning points are used, accelerating the processing time by approximately 10 times. Small position errors are also minimized.

Table 3. Developed Non-Line-of-Sight Active Systems Using Confocal and Non-Confocal Methods

Publication	Year	Scanning Type	Sensor and Method	Laser Type	Temporal Resolution	Reconstruction Method	Processing Time
[46]	2019	Confocal	SPAD	High-power laser (> 1W)	Long exposure (180 min)	F-k Migration	180 min (static), 1.2 s (dynamic)
[36]	2020	Non-confocal	32x32 SPAD camera, non-scanning 3D imaging	70 ps pulsed erbium fiber laser	55 ps (bin width)	Virtual wave field, FBP	3 seconds (ToF measurement)
[7]	2020	Non-confocal	Silicon SPAD, TCSPC TTTR Mode (Moving and Static Surfaces)	Onefive Katana 10 HP fiber laser	Picosecond	Phasor Field	Several minutes
[20]	2020	Non-confocal	SPAD, Discrete phasor field diffraction model	-	-	Fast Fourier Transform, Inverse diffraction	Shorter compared to similar methods
[47]	2021	Both confocal and non-confocal methods	28 SPAD pixels (4 pixels connected to one TCSPC channel)	Onefive Katana HP laser (532 nm, 35 ps pulse width)	85 ps FWHM (laser + SPAD), 30 ps (single pixel SPAD)	LCT, F-k Migration, Phasor Field Reconstruction	Non-confocal: 60 s, Confocal: 1800 s
[48]	2021	Confocal	SPAD, non-line-of-sight sensing with picosecond temporal resolution	Picosecond pulse laser	1.4 ps	LCT	50 ms scanning time per time bin
[49]	2023	Confocal	SPAD (PD-100-CTD), Archimedean spiral scanning	670 nm pulsed laser (ALPHALAS)	4 ps	Spherical regularization	10 times faster processing with 1/23 scanning points

These studies demonstrate that confocal and non-confocal systems can be diversified by using different geometrical layouts, optical arrangements, laser sources, and algorithms. This diversity offers significant technical advantages in terms of depth resolution, processing time, and signal quality in NLOS imaging. In short, positioning the source and detector along the same axis in confocal methods, as opposed to positioning them on different axes in non-confocal methods, plays a decisive role in performance and processing challenges. Table 3 provides an overview of studies on non-line-of-sight active systems developed using confocal and non-confocal methods.

2.2.3. Acoustic Methods

Acoustic methods in non-line-of-sight analysis are techniques that aim to obtain information by analyzing the reflection and return of sound waves from inaccessible areas. These techniques analyze the reflection and refraction of sound waves as they interact with objects, such as obstacles and walls, as shown in Figure 5, and detect hidden objects. Similar acoustic mechanisms for NLOS object classification have also been reported in the literature [50]. The return signals are collected via microphones, and information about the location and shape of the hidden object is extracted using reverse algorithms. Acoustic non-line-of-sight technology is utilized in various applications, including positioning and localization operations, sound source localization and direction determination, and the reconstruction of objects behind walls and obstacles. This method offers a significant advantage in visualizing hidden objects around obstacles by leveraging the physical properties of sound waves, particularly in environments where optical devices are unavailable or limited.

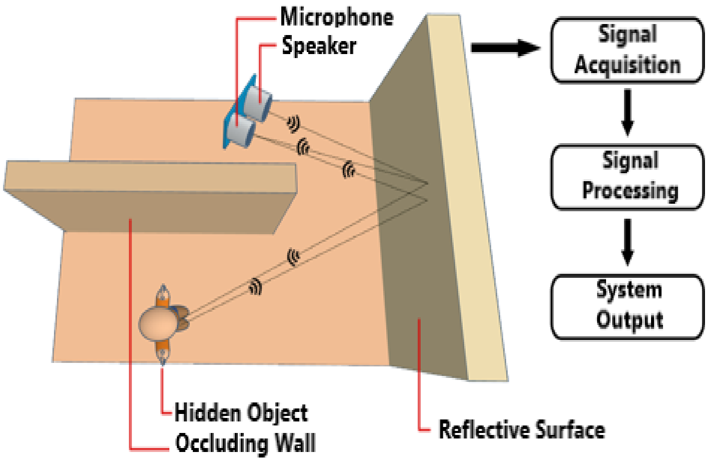

Figure 5. Acoustic NLOS Object Detection Process Schematically Illustrated for This Review

In their study [51], King et al. aimed to detect hidden objects in unseen corners by acoustic methods in addition to the image sensors already present in autonomous vehicles. In this direction, firstly, signals were divided into certain time windows and the energy density of frequency components in each time period was calculated. Components above the determined threshold value were marked as dominant sources for that time period. Then, the number of active sources in these regions and their directions were determined by criteria such as the differences between their energy densities. The GCC-PHAT algorithm was used for direction detection. GCC-PHAT performed signal arrival direction (DoA) estimations by using the first and second peaks. Finally, the estimated directions were grouped with the k-means algorithm, and the results were optimized. As a result of the study, it was demonstrated that the secondary peaks of the GCC-PHAT algorithm supported the primary DoA estimations, and dynamic DoA detection provided higher accuracy with measurements taken at different locations. This study demonstrates that the sound source can be successfully localized by combining data collected from multiple locations.

In their study [52], An et al. developed an approach that combines techniques such as acoustic ray tracing and TDoA-based methods with data obtained from sensors used on a robot to determine the location of the sound source. In this method, multiple sound data were collected via sensors placed on a robot. These data were analyzed in conjunction with sound directions, intensities, and representative frequencies using a TDoA-based approach. However, since it was not yet determined whether these directions were direct or indirect, the acoustic ray tracing method was applied. The robot used the SLAM (simultaneous localization and mapping) method to map its environment and determine its location. Then, rays were generated in the opposite direction of the acoustic rays reaching the sensors, and these rays moved forward by hitting a wall or an object and reflecting off it. The point where the two rays intersected was assumed to be the object location, but the Monte Carlo method was used for a more precise result. In this method, random particles were first selected, and then weights were assigned to the rays according to their proximity. When sufficient proximity was not achieved, low-weight particles were eliminated, and new samples were generated based on high-weight ones. This process continued until the generalized variance value fell below 0.01. Based on the results obtained, the region where the sound source is located was defined in an elliptical shape. The TurtleBot2 robot, Kinect sensor, laser scanner, and microphone array were used in the experiments. Applause sound with frequencies above 2 kHz was preferred as the sound source. Sound production was changed every 5 seconds, and it was observed that the distance errors averaged 0.72 m.

The aim of the study by Schulz et al. [53] is to detect the sounds of vehicles without line of sight by reflecting them off the surrounding surfaces using a microphone array mounted on a vehicle in real-world conditions. In this context, DoA analysis was used to determine the direction from which the sound signals came. DoA is determined by analyzing the arrival times or phase differences of sound waves between different sensors. For this purpose, the SRP-PHAT method was used to determine the direction from which the sound signals coming from multiple microphones on the vehicle came. The method calculates the direction from which the sound came by analyzing the time differences in arrival between microphone pairs. In this direction, the obtained DoA value is compared with a certain threshold value, and it is concluded that the approaching vehicle is coming from the left, front or right direction. The study achieved an accuracy rate of 92% in static conditions and 84% in dynamic conditions. In addition, vehicles were detected more than a second earlier compared to visual detection methods (Faster R-CNN).

The work of Lindell et al. [54] focuses on the use of acoustic signals for NLOS imaging of objects behind a wall. During the application, the sound signal sent through a speaker bounces off the object behind the wall and is detected by the microphones. In the experiment, a system consisting of 16 microphones and 16 loudspeakers is placed in a mechanical system that can move and change direction. Each speaker plays sound in turn, and a total of one second of sound is transmitted for each microphone for 0.0625 seconds. The hardware in the system comprises omnidirectional measurement microphones, automobile speakers, and acoustic interfaces, including microphone preamps and sound cards. The sound signals are converted into a digital format and synchronized with fiber-optic cables. Additionally, amplifiers are used to amplify the sound and transmit it to the speakers. The microphone and speaker array are mounted vertically to take measurements at 32 positions, with 1-meter intervals between each position. There is a foam barrier between the hidden object and the array. The transmitted audio signals are set to linear frequency chirp from 2 to 20 kHz with a period of 0.0625 seconds for each speaker. The sound level is about 80 dBSPL at 1 meter, and the maximum range of the system is 10.6 meters. Experimental results demonstrate that in acoustic non-line-of-sight analysis, objects are clearly reconstructed, and accurate 3D positioning is achieved. Furthermore, acoustic non-line-of-sight analysis exhibits superior performance with lower signal drop compared to optical non-line-of-sight analysis.

These studies demonstrate that acoustic signals provide high accuracy and flexibility in analyzing objects non-line-of-sight by leveraging their temporal, directional, and frequency properties; therefore, acoustic-based NLOS methods are a strong alternative to optical systems.

2.2.4. Artificial Intelligence Based Active Methods

One of the biggest challenges in non-line-of-sight analysis is correctly interpreting the return signals that reach the sensors by reflecting from surfaces and creating a visual representation. While traditional methods try to overcome this difficulty with nonlinear algorithms, physical and geometric modeling, successful studies have been carried out in this field with deep learning (DL) techniques in recent years. Artificial intelligence models trained on large datasets have significantly increased the solution potential by easily analyzing the complex structure of reflected signals. Deep learning models offer the advantage of high-accuracy reconstruction with shorter processing times and lower computational requirements. Increasing success with traditional physical modeling methods is a difficult process; on the other hand, deep learning methods can offer flexible and adaptable solutions by learning from data. Deep learning models obtained as a result of training can analyze signals in real time and provide instant location information or reconstruction. This provides a significant advantage, especially in real-time operations such as autonomous systems, search and rescue and security applications.

In their study [6], Lei et al. aimed to detect information in speckle patterns formed by objects non-line-of-sight on reflective walls under coherent illumination (laser-based phase-stable lighting) using deep learning techniques. Non-line-of-sight object recognition was performed from speckle patterns using low-cost devices such as CMOS sensors. The interpretation of speckle patterns is a challenging process, but deep learning techniques can extract the information required for object recognition. Both experimental and simulation results are examined in the study. Two separate data sets were used in the study. While the first 10,000 data points from the MNIST data set were used in both experiments and simulations, additional data from 12 human subjects, representing 10 different posture categories, were used in the experiments. In the preprocessing stage, speckle patterns were cropped, random noise was added to the speckle patterns to simulate distortions that may occur in a real environment, and then normalized to a range of 0 to 1. In the posture data set, the background was cleaned using DeepLab-v3 and converted to a gray level. Speckle patterns cropped to 200x200 dimensions were trained with the SimpleNet network. For human pose classification, random parts of 224x224 size from cropped images of 256x256 were trained using the ResNet-18 model. As a result, while an average recognition accuracy of 78.18% was obtained for human pose, this value was over 90% for the MNIST dataset.

Metzler et al. [55] aimed to develop a model to address the problem of signal-to-noise ratio (SNR) decrease with distance. In this direction, they utilized spectral estimation theory to mathematically model the attenuation of light power and generated synthetic data using a noise model based on speckle correlations. The data generated by passing the Berkeley Segmentation Dataset 500 (BSDS500) through the Canny edge detector was cropped to create images with sparse edges. A deep learning model (CNN) was created using the obtained data and combined with a deep inverse correlography application to mitigate noise resistance. They aim to overcome the difficulties of PR (Phase Retrieval) algorithms in low light conditions and to improve their performance. In this study, the U-Net architecture is utilized as a neural network and implemented using PyTorch. The batch size was set to 32, the learning rate was started with 0.002 and decreased to 0.000001, and training was performed for 400 epochs. The training using the ADAM optimization algorithm was completed in approximately one day. The results obtained were compared with traditional PR algorithms (HIO or Alt-Min), and more efficient calculations and stronger performance against various types of noise were obtained. In this way, images were obtained with a 1/8 second exposure time using a traditional CMOS detector and the shapes of small hidden objects at a distance of one meter were successfully reconstructed.

Isogawa et al. [56] presented how visual data should be represented and processed to estimate 3D poses from non-line-of-sight data and a different method of using machine learning for this purpose. Using the MoCap motion capture system, a confocal non-line-of-sight analysis method, and depth maps, 3D human poses are obtained from real subjects, and fake images are generated from them. Errors such as Poisson noise, time shift, and time blur are added to the fake images to make them closer to real-world data. Features are extracted from temporary images using the P2PSF Net network and prepared for use in the simulator. The network takes data at a resolution of 32x32x64 and increases the volume of this data to 128x64x64. The network contains 9 3D convolutional layers and 2 residual connections. The network corrects sensor errors and creates a reflection volume. This volume is converted into a 2D heat map using max pooling, and then meaningful features are extracted with ResNet-18. Thus, the necessary features for 3D human pose estimation are obtained. A humanoid control policy is created using a Markov Decision Process (MDP) with feature vectors extracted from images obtained using the P2PSF Net network. This policy network, trained with deep reinforcement learning techniques, calculates joint movements in the control policy in the physics simulator. The human pose estimation simulator is created using mathematical formulas that model physical processes in the human body and can make estimations based on images. MPJPE, Evel, Accl, and Ekey values obtained as a result of the study were measured as 96.1, 4.62, 4.33, and 0.173, respectively.

In their study [57], Zheng et al. aim to reconstruct images by training deep neural networks with speckle patterns of hidden objects illuminated by a plasma light source with high energy density and a broad spectrum. For this purpose, two different DNNs (Deep Neural Networks) are designed: DNN1, which eliminates aliasing and background effects caused by illumination and highlights the details of speckle patterns, and DNN2, which reconstructs the hidden object from the optimized patterns generated by DNN1. Two different datasets are used to train the network, which is designed similarly to a typical U-Net structure: the first group consists of single-digit handwritten images from the MNIST dataset, and the second group consists of three-digit handwritten images. From the images obtained as a result of training, RMSE and CC values are calculated as 47.446 and 0.867 for the first group, and 44.433 and 0.865 for the second group, respectively. These values show that the reconstruction performance of the proposed method is successful. In addition, tests were conducted by changing parameters such as wall type, object type, camera and object angle, and it was shown that the proposed method is also robust to changes.

The first study applying transformer-based approaches to NLOS analysis was introduced through a method called NLOST [58]. This model employs specially designed spatial-temporal self-attention and cross-attention mechanisms to capture local and global correlations in time-resolved ToF measurements. Shallow features extracted with physics-based priors are transformed into deep local and global representations via transformer-based encoders, which are then fused to reconstruct the 3D volume of hidden scenes. Experiments on simulated data demonstrate that the proposed method achieves superior accuracy compared to traditional physics-based approaches (FBP, LCT, FK and similar methods) and deep learning-based methods (U-Net, LFE, NeTF). Moreover, the model, trained only on synthetic data, successfully generalized to real-world measurements obtained from publicly available datasets and a custom-built experimental system. These findings show that NLOST provides high resolution, robustness to noise, and strong representational capability for real-world applications in NLOS analysis.

Shen et al. [59] introduce MARMOT, the first self-supervised masked autoencoder pretrained for NLOS reconstruction. Using a Scanning Pattern Mask (SPM) to randomly hide most transient measurements and a Transformer encoder–decoder to predict the missing signals, MARMOT learns transferable features that complete the transient field from sparse, irregular inputs. Pretrained on TransVerse (~500K 3D objects) with realistic SPAD noise, the model achieves superior performance in transient completion and 3D shape recovery, outperforming physics-based (LCT, f-k, and similar methods) and learning-based (NLOST) baselines on downstream tasks such as albedo, depth, and classification. Notably, MARMOT remains effective under 95% masking, highlighting its robustness to sparse sampling. Remaining limitations include the planar relay assumption, a focus on confocal setups, and the limited material complexity in the dataset.

These studies show that deep learning and machine learning based approaches provide high accuracy and flexibility in complex NLOS tasks such as speckle pattern analysis, pose estimation, and image reconstruction, thus making strong contributions to unseen scene analysis, overcoming the limitations of traditional methods. Table 4 provides an overview of these active system studies developed using AI-based methods.

Collectively, these studies demonstrate that AI-based approaches can significantly enhance NLOS imaging by providing high accuracy, robustness to noise, and the potential for real-time operation. However, each method also entails application-specific trade-offs: speckle-based recognition is low-cost but sensitive to environmental variability; noise-robust CNNs improve performance in low light conditions at the expense of computational cost; pose estimation frameworks are powerful for human-centric tasks but depend on complex motion capture data; plasma light driven DNNs achieve detailed reconstructions yet face scalability challenges; and transformer/self-supervised models such as NLOST and MARMOT set new accuracy benchmarks but rely heavily on synthetic datasets and specific system assumptions. This indicates that while deep learning expands the design space of NLOS imaging, tailoring architectures and training protocols to target environments remains an open research priority.

Table 4. Active NLOS Systems Developed Using Artificial Intelligence-Based Methods

Publication	Year	Dataset Used	Method and Technology	Defined Problem	Deep Learning Method	Results
[6]	2019	MNIST and human pose dataset	Single and multi-wall environments, reflected light, CMOS camera	Non-line-of-sight recognition, human pose detection	SimpleNet, ResNet-18	Experimental accuracy: 95%, simulation: 97%, MNIST accuracy: over 90%, human pose accuracy: 78.18%
[55]	2020	Berkeley Segmentation Dataset 500 (BSDS500)	Noise modelling, autocorrelation, low-light and noisy environments	Reconstruction in non-line-of-sight noisy scenes	U-net	Successfully reconstructed shapes of small hidden objects at 1m with 1/8 sec exposure time.
[56]	2020	Measured and synthetic data	Pulse laser, ToF sensor, MoCap, P2PSF Net, MDP	3D human pose estimation from transient images	P2PSF (CNN, Residual, ResNet-18), MDP, Deep RL	MPJPE: 96.1, Evel: 4.62, Accl: 4.33, Ekey: 0.173
[57]	2021	MNIST dataset	White light illumination, speckle patterns, two-step DNN method	Non-line-of-sight reconstruction, impact of varying parameters	Two-stage DNN similar to U-net	Good generalization across different wall and object types; Group 1 RMSE: 47.446, CC: 0.867; Group 2 RMSE: 44.433, CC: 0.865
[58]	2023	Synthetic data (ShapeNet Core motorcycles, other objects) + Real-world datasets	Transformer with spatial-temporal self- and cross-attention	Improving NLOS 3D reconstruction accuracy and generalization in complex scenes	Transformer (NLOST)	Outperformed physics-based (FBP, LCT, FK, RSD) and DL methods (U-Net, LFE, NeTF); robust generalization to real-world data
[59]	2025	TransVerse (~500K 3D objects, synthetic + SPAD noise)	Self-supervised masked autoencoder (Transformer encoder–decoder, Scanning Pattern Mask)	Completing sparse/irregular transient measurements for NLOS reconstruction	Transformer-based MAE (pretrained, feature transfer, finetuning)	Outperforms physics-based (LCT, f-k, etc.) and learning-based (NLOST); robust under 95% masking

2.3. Passive Methods

Passive non-line-of-sight analysis involves the process of signals such as light sources (daylight, ambient lighting, etc.) found naturally in the real environment, thermal radiation, and natural acoustic waves reaching the sensor by reflecting from objects or walls and analyzing them. In light-based passive imaging, scene information is obtained using only the reflections of existing ambient light or shadows, without the need for active illumination, such as a laser. The intensity, timing, and spectral properties of the signals are extracted from the reflection information measured with optical sensors. Thermal passive imaging is the analysis of infrared radiation emitted by living beings and objects in the environment using thermal cameras. Acoustic passive imaging is the analysis of the echoes of natural acoustic signals in the environment with microphones. Passive methods are low-cost since they do not require an external source; however, natural signals are generally weaker, and their analysis is more complicated due to scattered reflections.

Figure 6 presents a comparative workflow of traditional and artificial intelligence (AI)-based passive NLOS imaging methods. In both approaches, the process begins with scene preparation, where the hidden object is positioned outside the line of sight, and the camera is directed toward a surface capable of capturing indirect light reflections. This is followed by visual data collection, in which indirect light reflected from intermediate surfaces is recorded.

In traditional passive methods, the collected data undergoes physical modeling, where light propagation and reflection behaviors are mathematically formulated. Subsequently, an inverse problem solving stage infers the hidden scene's geometric or photometric features, leading to a reconstruction step that generates a visual representation of the obscured object. The final output is presented through visualization and interpretation.

Conversely, in AI-based passive methods, the visual data is first subjected to data preprocessing (e.g., normalization, filtering, or resizing), preparing it for the deep learning model. The AI model training phase enables the model to learn the relationship between reflection patterns and hidden objects using labeled datasets. During inference, the trained model predicts the shape or position of the hidden scene, which is then reconstructed into a visual output. This output, similar to the traditional method, is finally analyzed in the visualization and interpretation stage.

Figure 6 effectively illustrates the sequential flow and key distinctions between traditional and AI-driven passive NLOS imaging approaches.

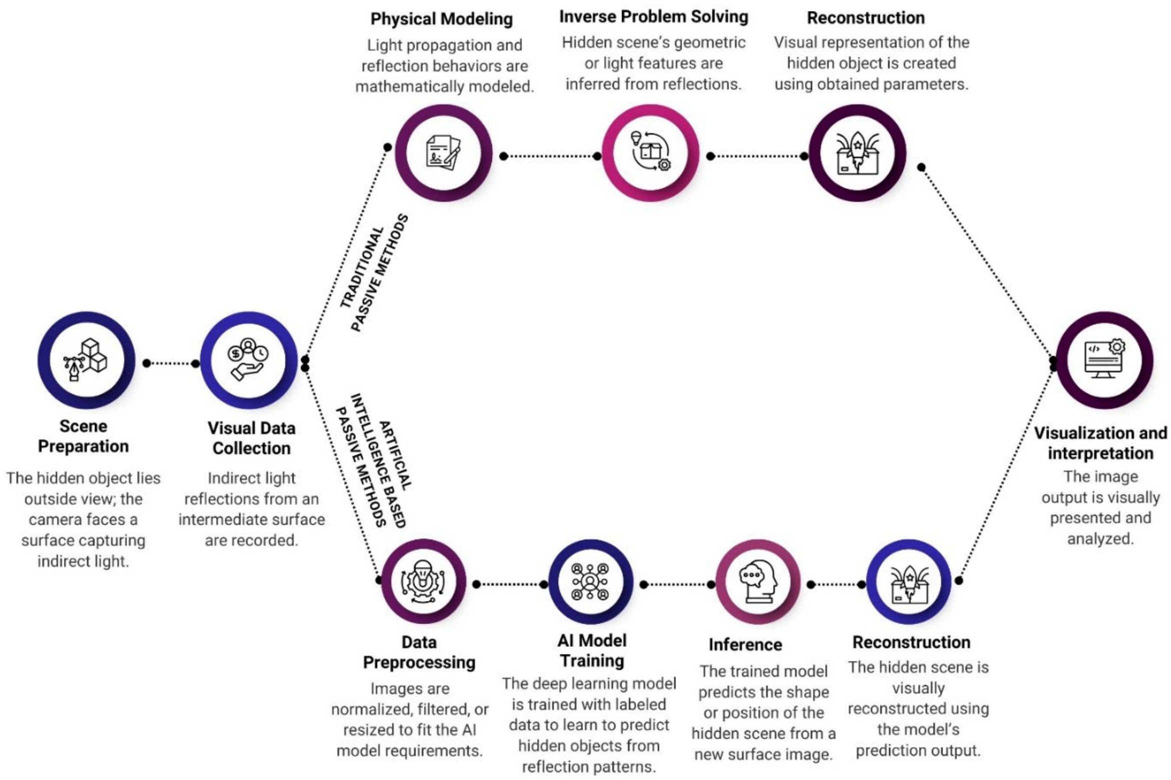

Figure 6. Comparative Workflow of Traditional and AI-Based Passive NLOS Imaging Methods

2.3.1. Traditional Passive Methods

The work of Bouman et al. [60] aims to use vertical structures, such as wall edges, as a passive camera to detect hidden scenes. The integral of the hidden area light is reflected on the ground near these structures. By analyzing these shadows and subtle changes in the light gradient on the ground, 1D video samples are created, and the number of people and motion characteristics in the hidden scene are determined. In addition, it is explained how to find 3D locations of moving objects by using adjacent wall edges, such as doors, as a stereo camera. The work evaluates experimental results for different indoor and outdoor environments, including flooring types such as tiles, concrete, wood, carpet, brick, etc.

In their study [61], Boger-Lombard and Katz aimed to detect the location of the hidden light source behind the barrier by measuring the time delay (ToF) of the white light leaving the strong scattering barriers. In the study, a randomly emitted white light with a wide spectrum is used. The time difference between the light leaving the source and returning from the barrier is measured by cross-correlation. Low coherence spectral patterns are created using a conventional camera, and the location and motion of the light source are estimated by these interference patterns.

Beckus et al. [62] present a method that reconstructs the hidden scene using the spatial coherence information of the hidden object signals reflected from a diffuse wall. The spatial coherence information is obtained using the 4D spatial coherence function, and the spatial properties and intensity of the reflected light are combined at different scales through a multi-criteria convex optimization problem method. The Alternating Direction Method of Multipliers algorithm is used to optimize multiple parameters to provide more accurate and faster imaging. Multimodal data fusion is achieved by combining data types such as imaging data and light intensity data. In the study, the performance is measured by the Mean Square Error (MSE) metric calculated for different noise levels (nI) and regularization parameter (κ). For example, while the MSE value obtained for the value of $\kappa = 0$ and the noise level of $nI = 0$ is 0.04, this value increases to 0.09 when the value of $\kappa = 1.00$ is reached. At higher noise levels ($nI = 1$), the MSE is found to be 3.019 ± 1.817 for $\kappa = 0$, while this value decreases to 0.096 ± 0.006 with the value of $\kappa = 1.00$. This shows that the regularization parameter (κ) reduces the sensitivity to noise and allows lower MSE values to be achieved.

In addition to optical-based passive methods, passive approaches using acoustic waves have also been proposed recently. Tao et al. [63] developed the reflection-enhanced underwater acoustic passive localization (RUAPL) method and evaluated the spatial diversity of all multipaths. In this study, multipath arrival structure (MAS) extraction based on spatial time-frequency analysis was performed with accuracy close to the Cramer–Rao lower bound (CRLB). Following this, LOS and NLOS paths were weighted differently using HDOP-based reliability measurement. Simulations and lake test results showed that RUAPL reduced localization error by up to 128% compared to traditional hybrid methods. Similar to how traditional passive optical approaches (e.g., shadow or speckle-based methods) exploit ambient or scattered fields without active sources, acoustic-based systems can also be extended within this passive framework by relying on naturally reflected sound waves. These results demonstrate that passive acoustic methods also have significant potential in NLOS scene analysis.

Passive NLOS imaging also utilizes the infrared/thermal band. Chen et al.'s approach [64], called NLOS-LTMD, performs passive NLOS pedestrian imaging by capturing reflection signals at long distances using an infrared camera. The study decomposes the light transport matrix into illumination and reflection components and develops an IO-Transformer-based deep learning model to mitigate low SNR issues. Furthermore, the first infrared passive NLOS dataset (IF-NLOS), consisting of 33,000 images, is presented, and experiments demonstrate that the method exhibits higher accuracy and robustness than existing passive NLOS techniques.

Passive NLOS methods are not limited to optical or thermal signals; wireless communication signals can also be used for this purpose. One approach proposed in the literature [65] focused on modeling the intrusion detection margin using Channel State Information (CSI) data in indoor NLOS scenarios. The study derived a model that describes the impact of human movements on the NLOS wireless channel, and based on this model, the CSI-based intrusion detection margin was theoretically explained. Furthermore, a practical margin estimation system was developed and validated in real-world experiments, demonstrating that passive RF signals can be effectively utilized in security-focused NLOS applications.

These studies demonstrate that, in addition to optically based passive approaches such as ambient light-based shadow analysis, time delay measurements, and spatial coherence, methods based on acoustic signals, techniques using infrared/thermal radiation, and solutions based on wireless RF signals also enable successful analysis of NLOS scenes with low hardware complexity. Thus, passive methods are not limited to optics but also gain interdisciplinary diversity by utilizing different wavelengths. Table 5 provides an overview of systems developed using optical, acoustic, thermal, and RF-based passive methods.

Table 5. NLOS Systems Developed Using Traditional Passive Methods

Publication	**Year**	**Method**	**Working Principle**	**Signal Type**	**Advantages**	**Limitations**	**Technological Requirements**
[66]	2012	Shadow-based imaging	Analyzing shadows formed by ambient light passing through occluders to infer outdoor scenes and light sources	Ambient light (natural)	Low cost, no energy required	Depends on ambient light, low resolution	Consumer-grade cameras
[60]	2017	1D video capture around corners	Light blocked by vertical wall edges creates angular shadows. These are converted into 1D videos showing motion around corners.	Ambient light (penumbra shadows)	Easy measurement with regular cameras; can detect low-contrast signals; stereo view with open doors	Sensitive to lighting; detects only corner-adjacent motion; not suitable for wide-area imaging	Consumer-grade cameras

Table 5. (Continued)

[61]	2019	Passive Time-of-Flight (ToF) imaging	Uses cross-correlation of white light ToF delays to locate hidden light sources behind barriers	White light (LED, halogen)	Accurate, easy to set up, and provides illumination position info	Limited range, depends on reflectivity, and is affected by ambient light	LED/halogen light sources, diffusers, reflective walls, CMOS camera, lens, aperture masks, ToF system
[62]	2019	Passive NLOS using spatial coherence	Reconstructs object profiles by combining spatial coherence and intensity data from different modalities	Spatial coherence and intensity	Robust to noise, improved accuracy via multimodal data	Needs filtering with sample weighting in high noise	Cameras, optical sensors, depth info, regularization, algorithm tuning
[65]	2023	CSI-based Passive Intrusion Detection	Modeling the impact of human movement on NLOS wireless channels to estimate CSI-based intrusion detection bound; validated through real-world experiments.	RF / Wireless (CSI)	Effective for security-oriented applications; validated in real-world environments; low hardware complexity	Scenario-dependent model; limited generalizability across different RF environments	Wireless CSI measurement systems, RF receivers, data processing algorithms
[63]	2024	RUAPL (Reflection-enabled Underwater Acoustic Passive Localization)	Extraction of multipath arrival structure (MAS) and HDOP-weighted localization; assigning different weights to LOS and NLOS paths	Acoustic	Near-CRLB accuracy; up to 128% error reduction; validated by both simulation and lake experiments; high performance under low noise	Limited to underwater environments; dependent on a small aperture array; relatively high computational cost	Underwater acoustic arrays, time-frequency analysis algorithms, and HDOP-based computation
[64]	2025	NLOS-LTMD (Light Transport Matrix Decomposition)	Decomposition of the light transport matrix into illumination and reflection components; an IO-Transformer-based deep learning model to mitigate low SNR	Infrared / Thermal (IR)	First large-scale IR passive dataset (33,000 images); improved SNR with deep learning; higher accuracy and robustness than existing passive methods	Requires large datasets; dependent on deep learning models	Infrared cameras, a large-scale dataset, and high computational power for deep learning

2.3.2. Artificial Intelligence Based Passive Methods

In their study [8], He et al. aimed to detect the reconstruction and location of hidden objects in a scene at high speed without using an active light source, and designed an approach that incorporates deep learning techniques. After testing the neural network models for use with the MNIST dataset in simulation environments created with Blender software, studies were conducted in two different scenes. In the first scene, a working principle similar to a perforated camera was applied by passing the signals emitted from self-luminous objects through the window hole and processing the penumbra information on the reflective surface. In the second scene, reflections were created by shooting 15-second videos with ordinary ambient lighting from a moving mannequin with diffuse reflection properties. A chair was placed between the mannequin and the reflective surface to simulate the blocking of reflections by furniture. The real images and reflection images taken from the cameras were processed with deep learning techniques, and image and location estimation were performed. An architecture called DCIGN, based on the encoder and decoder architecture, was developed to reconstruct the images. The Encoder part consists of convolutional layers and pooling layers to extract and abstract features from input images. In contrast, the Decoder part reconstructs the output images by taking samples from feature distributions with deconvolutional layers. On the other hand, a Convolutional Neural Network (CNN) with a simpler architecture consisting of five convolutional layers and three fully connected layers was used for positioning tasks. As a result of the experimental tests, it is shown that reconstruction is performed in 3 ms with an error of approximately 1.83% and positioning is performed with an error of approximately 0.5 cm in x-y coordinates.

Wang et al. [67] proposed a deep learning network called SPIR-Net to extract the location and image of objects by analyzing speckle patterns. In this method, a holographic image is generated using laser light and a spatial light modulator (SLM). After the image is projected onto the wall, the location and image of the object are estimated by analyzing the speckle patterns recorded by the CCD camera. SPIR-Net works with the combination of LeNet, U-Net and cGAN networks. LeNet is specialized to perform the localization task. The modified LeNet avoids the loss of localization accuracy by using convolution layers instead of max pooling layers and increases the depth of the network. In this way, strong classification performance is achieved with low computational cost. U-Net performs pixel-level segmentation of images to identify the shape of objects more precisely. U-Net prevents the loss of low-level features by using "skip connections" between down-sampling and up-sampling stages and obtains more accurate results. While down-sampling reduces the image size and summarizes the general structure, up-sampling brings back higher resolution details, which provides clearer and more accurate segmentation in non-line-of-sight analysis. cGANs, on the other hand, help reconstruct the image of the object more realistically by adding a discriminator to the SPIR-Net model. This discriminator compares the images generated by SPIR-Net with the real images and determines the differences between the fake and real images. During the training process, SPIR-Net is constantly improved according to the feedback from the discriminator and tries to produce more realistic images. This interaction encourages SPIR-Net to create more accurate and high-quality object representations, thus increasing the quality of the images. In the study, MNIST and Quick Draw datasets were selected as the datasets and a holographic image was created via SLM. A 256x256 pixel section of the speckle patterns obtained from these images was used for training the network. SPIR-Net can recognize the longitudinal position of an object accurately even when the range is very small (low resolution). However, the recognition accuracy of positions in the transverse plane is low.

In their study [68], Wang et al. present a passive method that tracks the motion of a hidden object by analyzing only wall reflections without using expensive equipment. The motion information between the previous state and the current state is obtained using difference frames. The PAC-Net neural network was developed to obtain the dynamic and static properties of the motion from the difference frames. The network consists of Propagation-Cell modules, which process motion information across difference frames, and Calibration-Cell modules, which ensure the continuity of the motion path by calibrating with the raw frames. Propagation-Cell learns the relationship between the previous state of the object and the new observations from the difference frames using an RNN cell. Calibration-Cell processes the position information from the raw frames using the ResNet-18 network and makes the new state information updated by Propagation-Cell more precise. The new hidden state, formed as a result of these processes, is given as input to an MLP, and the position and speed of the object are estimated. Since the model does not have any scene information at the beginning, it may cause errors. Therefore, the hidden state of the model is initialized from a meaningful distribution with Warm-up PAC-Cell, preparing it for the tracking process. Both real environment data and synthetic data were used as data sets, and the first comprehensive passive non-line-of-sight tracking data set (NLOS-Track) was created with these data. Experimental results showed that the developed model can track a person moving in the hidden scene with centimeter-level precision. In addition, the RMS values of tracking errors and speed component (RMSx and RMSv) in the x-trajectory, the area between the estimated and real image (Area), general similarity error (DTW) and local error (PCM) values were measured as 1.37, 1.28, 0.0044, 0.33 and 0.30 with real data, and 8.78, 1.79, 0.084, 4.39 and 2.268 with synthetic data, respectively.

In the study by Wang et al. [3], a physics-based imaging prototype was presented from light spots emitted by moving objects using sensors such as event cameras. Unlike traditional frame-based cameras, event cameras only record the brightness changes of pixels at moving points of light. Event cameras calculate brightness changes in pixels in a logarithmic form and record two types of events: brightness increases or decreases, according to a certain threshold value. Time surface (TS) calculation combines the time and space features of events recorded by the camera and expresses the data with a map. The proposed model combines the event-based data obtained from event cameras with frame-based data obtained from traditional cameras to create a dataset. Certain physical laws are incorporated into the model to ensure that it produces more accurate results. The model is an end-to-end deep learning method consisting of encoder and decoder layers. While meaningful features are extracted from sensor data in the encoder layer, the hidden scene is reconstructed from these features in the decoder layer. In addition, the integration of physical laws provides a more advanced decision mechanism. The network trained with simulation data is then trained with real data and fine-tuned. As a result of the study, the measurement results made with frame-based, event-based and fusion of both data were compared, and it was observed that the studies made with data taken from event cameras gave better results. According to an example result, the results of PSNR/dB and LPIPS measurements made with event cameras were measured as 20.64 and 0.045 for close group objects, respectively, and 19.88 and 0.051 for far group objects, respectively.

Li and Zhang [69] present USEEN (Unseen Scene Encoding Enhancement Network), a passive NLOS imaging framework enabling real-time reconstruction of room-scale hidden scenes. Leveraging only diffuse reflections captured by consumer-grade CMOS cameras, the method offers a cost-effective and privacy-preserving solution for smart home security. The architecture combines U-Net with a conditional GAN, enhanced with multi-scale Inception blocks to balance fine detail recovery and global consistency. Experimental results demonstrate real-time performance (12.2 ms per estimation) with validated PSNR (19.21/15.86/13.62 dB) and SSIM (0.83/0.68/0.59) scores. Compared with DCIGN and pix2pix, USEEN achieves clearer and more detailed reconstructions while showing robustness to ambient light interference. These findings highlight its strong potential for smart home security, indoor navigation, and privacy-preserving monitoring applications.

These studies have shown that AI-based passive approaches enable fast, precise, and realistic reconstruction of objects non-line-of-sight using only ambient light or reflection patterns, and offer flexibility and accuracy beyond the reach of traditional methods. Table 6 provides an overview of passive NLOS systems developed using AI-based methods.

Collectively, AI-based passive NLOS methods provide promising alternatives by exploiting ambient light and reflection patterns without active illumination. However, each approach presents trade offs that determine its practical suitability: pinhole like or speckle-based reconstructions are computationally efficient but sensitive to noise and scene complexity; networks such as SPIR-Net achieve accurate segmentation yet suffer from low transverse resolution; motion tracking frameworks like PAC-Net offer centimeter level accuracy but require warm up phases to mitigate initialization errors; event camera systems are robust to fast dynamics but rely on specialized sensors; and USEEN demonstrates real time capability with consumer grade devices but has not yet been extensively validated across diverse lighting and material conditions. These comparisons suggest that while passive methods broaden accessibility and reduce cost, their robustness and generalization to complex real world scenarios remain open challenges for future research.

3. Evaluation Metrics

Evaluation metrics are analysis methods used to measure the accuracy, quality and efficiency of the system. There is no single evaluation metric used as a standard in non-line-of-sight analysis systems. In the studies, success has been addressed from many different perspectives, such as the similarity between the obtained output and the reference image, the tolerance of the signals to noise, location errors and estimation accuracy. The analysis of the studies conducted NLOS has been carried out according to different evaluation metrics such as SNR [56], PSNR [3], Structural Similarity Index (SSIM), Root Mean Square Error (RMSE) [60], Correlation Coefficient (CC) [60], Full Width at Half Maximum (FWHM) [48], Mean Per Joint Position Error (MPJPE), Keypoint Error (Ekey), Velocity Error (Evel) and Average Acceleration (Aaccl) [57], Learned Perceptual Image Patch Similarity (LIPS) [3] and accuracy - loss [6]. The Signal-to-Noise Ratio (SNR) represents the ratio of signal strength to noise in a measurement, and a high SNR value indicates that the signal is dominant over the noise, making it easily perceptible. The Peak Signal-to-Noise Ratio (PSNR) evaluates the similarity between two images by measuring the numerical errors at the pixel level between the reconstruction output and the reference image. A high PSNR value indicates that the image is close to the original, but does not always provide an accurate impression of visual perception. The Structural Similarity Index (SSIM) evaluates the structural components of images, such as brightness, contrast, and texture, in a manner similar to the human visual system, and takes values between 0 and 1; values close to 1 indicate high visual similarity. Root Mean Square Error (RMSE) gives the overall error amount of the model by taking the square root of the average of the squares of the differences between the estimated and actual values. Still, it does not provide direct information about visual quality or structural similarity. The Correlation Coefficient (CC) evaluates the accuracy of the model by measuring the linear relationship between the original and reconstructed images. Values close to +1 indicate high similarity, while values close to -1 indicate contrast.

Table 6. Passive NLOS Systems Developed Using Artificial Intelligence-Based Methods

Publication	Year	Dataset	Method & Technology	Problem Addressed	Deep Learning Method	Results
[8]	2022	MNIST (for simulation), experimental data	Standard RGB camera, ordinary light source, penumbra analysis	Real-time NLOS imaging and localization of a hidden target	DCIGN (encoder-decoder style), CNN	3 ms reconstruction with 1.83% error, ~0.5 cm localization error in x and y coordinates
[67]	2023	MNIST and Quick Draw	Laser light, SLM,	Image and position	SPIR-Net (modified	Accurate longitudinal localization, even at low resolution; lower

Table 6. (Continued)

			holography, reflection, CCD camera, speckle analysis, SPIR-Net	estimation in NLOS scenes	LeNet, U-Net, cGAN)	accuracy in transverse positioning
[68]	2023	NLOS-Track	Laser light, motion detection, CCD camera, dynamic timing, tracking, C-Cell & P-Cell model	Motion tracking from passive video streams	PAC-Net (RNN, ResNet-18, MLP)	Real data: RMSx: 1.37, RMSv: 1.28, Area: 0.0044, DTW: 0.33, PCM: 0.30; Synthetic data: RMSx: 8.78, RMSv: 1.79, Area: 0.084, DTW: 4.39, PCM: 2.268
[3]	2024	Simulated and experimental data	Event- and frame-based sensor data, physically embedded model	Data scarcity, overfitting, and accurate modeling of physical transmission	End-to-end deep learning (encoder-decoder), physics-guided modeling	Accurate NLOS tracking of moving objects; near group: PSNR 20.64 dB, LPIPS 0.045; far group: PSNR 19.88 dB, LPIPS 0.051
[69]	2024	Room-scale dataset (diffuse reflections, CMOS cameras, 11 participants)	Passive NLOS, diffuse reflection, CMOS cameras, USEEN (CNN + conditional GAN with Inception blocks)	Real-time reconstruction of human activities in room-scale hidden scenes under ambient light	USEEN (U-Net + CGAN)	12.2 ms per estimation; PSNR: 19.21/15.86/13.62 dB; SSIM: 0.83/0.68/0.59; clearer and more robust than DCIGN and pix2pix

Full Width at Half Maximum (FWHM) reflects the relationship between resolution and accuracy by measuring the width of a signal up to half of its peak height; a lower FWHM value results in sharper images. In the context of NLOS pose estimation studies, metrics such as Mean Per-Joint Position Error (MPJPE), Keypoint Error (Ekey), Velocity Error (Evel), and Average Acceleration (Aaccl) can be used to evaluate reconstruction accuracy and motion smoothness. MPJPE measures the difference between the estimated and actual positions of each joint; Ekey evaluates the accuracy of 2D keypoints; Evel evaluates the differences in velocity sequences; and Aaccl evaluates the smoothness of motion transitions. In visual quality assessment, the Learned Perceptual Image Patch Similarity (LPIPS) metric evaluates perceptual similarity using pre-trained neural networks, unlike traditional PSNR and SSIM metrics, and more realistically reflects the differences perceived by the human eye. Finally, accuracy and loss metrics are used to measure the success of deep learning-based NLOS prediction models, with high accuracy and low loss values revealing the effectiveness of the model.

4. New Scenes and Scenarios in NLOS Analysis

Non-line-of-sight analysis allows for a variety of scenes and scenarios. New scenarios and scenes present different challenges for the technology to implement and research to evaluate and improve model performance. Scenarios that include different lighting conditions, surface materials, object geometries, and dynamic motions are used to test the generalization capabilities of non-line-of-sight algorithms. For example, the algorithms need to accurately reconstruct both moving and static objects in complex and dynamic scenes. New scenarios can also be used to evaluate how systems perform under conditions of low SNR, environmental noise, or insufficient data. This variety makes non-line-of-sight analysing technologies more robust and flexible for real-world applications.

The basic NLOS scenarios are divided into four basic categories as shown in Figure 7: passive NLOS detection with natural ambient light sources as shown in Figure 7 (a); passive NLOS analysis with natural light reflected by objects as shown in

Figure 7 (b); active NLOS analysis where the light source and the sensor focus on different points as shown in Figure 7 (c); and confocal active NLOS analysis where the light source and the sensor focus on the same point as shown in Figure 7 (d).

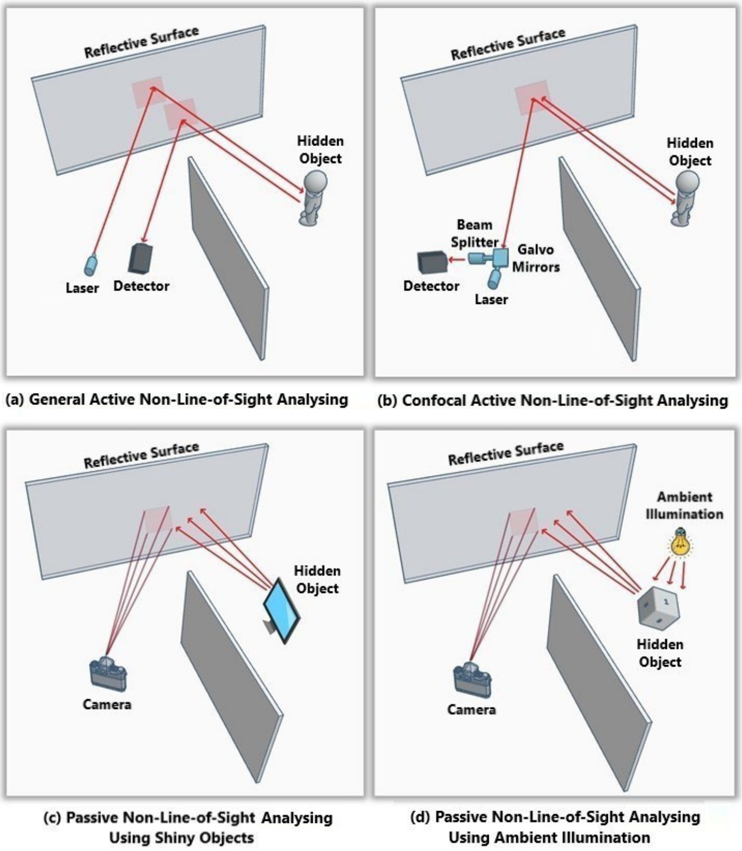

Figure 7. Basic Non-Line-of-Sight Analysis Scenes

NLOS scenarios are not limited to these. Many studies have been conducted by changing parameters such as the reflection pattern of the ambient light, the reflective wall, and the properties of the hidden object. Each of these studies has diversified the NLOS scenarios. In a study that sheds light on new scenarios, in contrast to traditional NLOS scenarios, instead of one obstacle wall, two obstacle walls were used to increase the number of reflections, as shown in Figure 8 (a). In this way, the traditional three-hop NLOS analysis approach was moved beyond the scope of the five-hop interaction scenario of light. In some studies [6, 7], movable reflective surfaces were used, as shown in Figure 8 (b). Another study [8] considered scenarios where reflections were restricted by placing obstacles between the hidden object and the reflective surface, as shown in Figure 8 (c). In another example [8], a scenario was examined by covering the hidden scene with a perforated screen, as shown in Figure 8 (d), simulating the operation of a pinhole camera. In addition, there are studies examining the pose of the object in the hidden scene [6, 8], [57] and its motion [45, 61] when the object is moving, as well as studies examining the measurement differences when the environment or reflective surface in the hidden scene has different materials [60, 61].

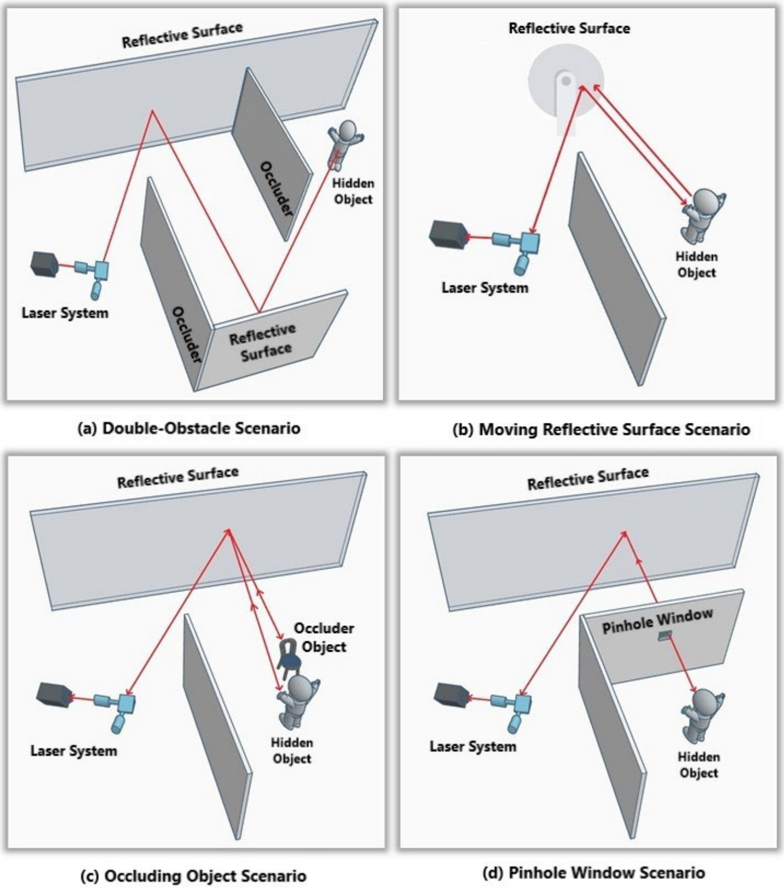

Figure 8. New Non-Line-of-Sight Analysis Scenes

Data fusion in non-line-of-sight analysing is an approach that aims to obtain more reliable and comprehensive outputs by integrating data obtained from different sensors. In acoustic sensing systems, signal propagation can significantly affect performance, especially in noise-sensitive environments; in such environments, NLOS sensing methods based on passive sound sources usually do not provide sufficient results. For this reason, the simultaneous use of acoustic and laser-based systems can provide more efficient results. Since acoustic systems can be negatively affected by ambient noise, data fusion with laser data can help overcome this problem. The aim of data fusion is to increase the accuracy of learning algorithms by combining data from sensors. For example, Schlosser and his team tried to solve the pedestrian detection problem by blending laser (LIDAR) data with image data [70]. Such studies show that data fusion has the capacity to create more accurate and detailed scene models. The combination of acoustic and laser signals is particularly useful when data is obtained with different sampling rates and measurement units [71, 72]. Although the integration of laser and audio data is limited in the literature, this combination can provide quite effective results. Combining laser and audio data can play an important role in detecting targets and providing more precise positioning.

5. Conclusion

This study aims to provide a comprehensive and in-depth analysis of the current state of non-line-of-sight (NLOS) sensing technologies by systematically examining the methodologies employed, the technological advancements made, and the critical challenges that continue to hinder their broader application. NLOS technology, which enables the detection and imaging of objects and individuals that are obscured from direct line of sight, represents a transformative innovation in the field of sensing and perception. It has garnered significant interest due to its wide range of potential applications across several high-impact domains. In particular, the ability to locate survivors trapped under rubble in post-disaster scenarios, to monitor concealed areas for security purposes, and to support autonomous navigation and situational awareness in robotic systems underscores the vital importance of further developing and refining these technologies. Despite these promising capabilities, the widespread implementation of NLOS systems remains constrained by several technical barriers. These include limitations in spatial and temporal resolution, the complexity and computational cost of processing large volumes of high-dimensional data, and the challenges posed by multipath propagation effects in both optical and acoustic domains, which can distort signal interpretation and reduce detection accuracy.

Within the scope of this study, a broad spectrum of methods and system architectures has been systematically reviewed and evaluated. These encompass both active and passive NLOS imaging systems, a variety of sensor technologies including

Single-Photon Avalanche Diode (SPAD) arrays and CCD/CMOS sensors, as well as imaging techniques based on confocal and non-confocal configurations. Each approach presents unique advantages and limitations depending on the application context and environmental conditions. For instance, SPAD-based systems offer exceptionally high temporal resolution, enabling precise time-of-flight measurements critical for 3D reconstruction tasks. In contrast, CCD/CMOS-based systems provide a more cost-effective alternative while maintaining adequate performance for processing speckle pattern data. Confocal systems are capable of achieving high spatial accuracy. They are particularly well-suited to environments with limited scattering, whereas non-confocal systems stand out for their reduced hardware complexity and flexibility in system design. In addition, acoustic-based NLOS systems, which rely on the propagation of sound waves rather than light, demonstrate effectiveness in scenarios where optical visibility is severely restricted, such as through walls or in dusty, dark environments. Artificial intelligence-driven models, particularly those employing deep learning algorithms, have also emerged as powerful tools for improving detection accuracy, enabling real-time inference, and enhancing generalization across different scene configurations.

To enable a fair and systematic comparison of different NLOS methods, a wide range of evaluation metrics has been employed. Classical image quality and error assessment measures such as Signal-to-Noise Ratio (SNR), Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), Root Mean Square Error (RMSE), and Correlation Coefficient (CC) are commonly used to quantify reconstruction fidelity. Alongside these, newer perceptual and task-specific metrics such as Learned Perceptual Image Patch Similarity (LPIPS), Mean Per Joint Position Error (MPJPE), and accuracy-loss have been introduced to capture both objective accuracy and perceptual quality, especially in systems that integrate machine learning components. Furthermore, the development of novel experimental scenarios and synthetic test environments has become a crucial aspect of performance validation. These include complex setups featuring multiple occluding walls, dynamic and non-Lambertian surfaces, low-reflectivity materials, and perforated barriers, each designed to simulate real-world challenges more effectively. Data fusion approaches, which combine multiple sensor modalities, material responses, and lighting conditions, also play a significant role in enhancing system robustness, expanding operational versatility, and improving the generalizability of results across diverse use cases.

While transformer and self-supervised approaches provide remarkable improvements in reconstruction accuracy, they often rely heavily on large-scale synthetic datasets. They may fail to generalize adequately in diverse, irregular environments. Similarly, passive methods hold promise for privacy-preserving and low-cost applications, but their sensitivity to ambient light and complex scattering limits broader deployment. A persistent challenge across both active and passive paradigms is balancing imaging quality, computational efficiency, and hardware feasibility. Bridging this gap requires not only algorithmic innovations but also joint design with sensor hardware, integration of data fusion strategies, and adoption of real-world validation protocols. In this context, combining laser and acoustic data holds strong potential to enhance detection performance under low-visibility or complex environmental conditions. Furthermore, developing more diverse and realistic datasets that cover a wide range of object types and scenario settings is critical for improving the adaptability and reliability of AI-based models. Therefore, while the techniques reviewed here mark important milestones, achieving scalable, robust, and widely adoptable NLOS systems demands a holistic integration of theoretical, algorithmic, data-driven, and engineering perspectives.

Future research should move beyond general improvements and focus on more targeted directions. In particular, the development of low-cost and energy-efficient NLOS systems will be critical for enabling widespread adoption in consumer and public safety applications. The integration of NLOS imaging into autonomous vehicles and robotic platforms represents another high-impact avenue, where robust perception in dynamic environments is essential. Moreover, advances in multimodal fusion approaches (e.g., combining laser and acoustic sensing) and the creation of diverse, real-world datasets will be necessary to improve generalization across different object types, materials, and environmental conditions. Such targeted efforts will accelerate the transition of NLOS imaging from controlled laboratory demonstrations to scalable, practical systems for everyday use.

Article Information Form

Authors Contributions: Conceptualization, S.Ç.; methodology, S.Ç.; resources, S.Ç.; writing-original draft preparation, S.Ç.; writing-review and editing, S.Ç. and İ.T.; visualization, S.Ç.; supervision, İ.T.; project administration, İ.T. All authors have read and agreed to the published version of the manuscript.

Conflict of Interest Notice: The authors declare that there is no conflict of interest regarding the publication of this paper.

Support/Supporting Organizations: This study was supported by the Scientific and Technological Research Council of Turkey (TUBITAK) under the Grant Number 123E386. The authors thank TUBITAK for their support.

Ethical Approval: It is declared that during the preparation process of this study, scientific and ethical principles were adhered to, and all studies cited are listed in the bibliography.

Artificial Intelligence Statement: During the preparation of this manuscript, the authors used ChatGPT (OpenAI) for language polishing and structural editing. No AI tool was used for data generation, analysis, or scientific interpretation. The authors reviewed and edited all outputs and take full responsibility for the content of this publication.

Plagiarism Statement: This article has been scanned by iThenticate™.

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